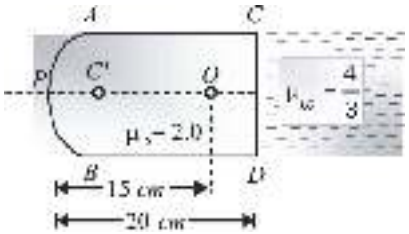
PHYSICS

1) The slab of a material of refractive index 2 shown in figure has curved surface APB of radius of curvature 10 cm and a plane surface CD. On the left of APB is air and on the right of CD is water with refractive indices as given in figure. An object O is placed at a distance of 15 cm from pole P as shown. The distance of the final image of O from P, as viewed from the left is



(1) 20 cm

(2) 30 cm

(3) 40 cm

(4) 50 cm

2) A copper calorimeter of mass 150 g contains 250 g of water at 20°C. Steam from boiling water in flask at 100°C is passed into calorimeter, till its temperature rises to 75°C. After condensation, the total mass of calorimeter and water is found to be 425g. Assuming no heat loss to surroundings, then calculate value of latent heat of vaporisation of water in cal g^{-1} is :
(Given : specific heat of copper = 0.1 cal $g^{-1}°C^{-1}$)

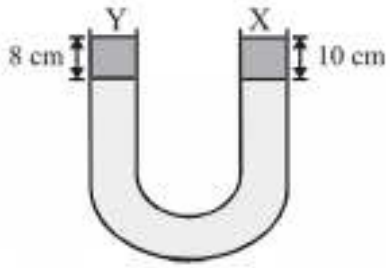
(1) 525

(2) 535

(3) 545

(4) 558

3) A liquid X of density 3.36 g/cm^3 is poured in a U-tube, which contains Hg. Another liquid Y is poured in left arm with height 8 cm. Upper levels of X and Y are same. What is density of Y ?



(Density of Hg = 13.6×10^3 kg/m^3)

(1) 0.8 gm/cc

(2) 1.2 gm/cc

(3) 1.4 gm/cc

(4) 1.6 gm/cc

4) A particle is rotating with constant angular acceleration on a circular track. If its angular velocity changes from 20π rad/sec to 40π rad/sec in 10 second, what are the number of revolutions that the particle has completed during this time ?

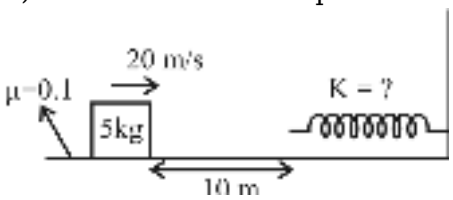
(1) 100

(2) 150

(3) 250

(4) 1000

5) If the maximum compression in the spring is 50 cm then find the value of K($g = 10\ m/s^2$) :-



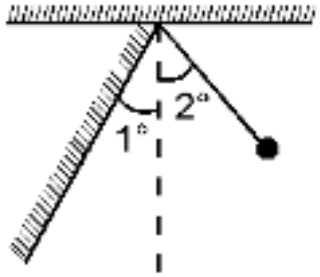
(1) $7580\frac{N}{m}$

(2) $7600\frac{N}{m}$

(3) $8420\frac{N}{m}$

(4) None of these

6) A simple pendulum of length 1 m is allowed to oscillate with amplitude 2°. It collides elastically with a wall inclined at 1° to the vertical. Its time period will be (use $g = \pi^2$) :-



(1) 2/3 sec

(2) 4/3 sec

(3) 2 sec

(4) none of these

7) **Assertion (A) :-** The drift speed of electrons in metals is small (in the order of a few mm/s) and the charge of an electron is also very small ($= 1.6 \times 10^{-19}C$), yet we can obtain a large current in a metal.
Reason (R) :- At room temperature, the thermal speed of electron is very high (about 10^8 times the drift speed).

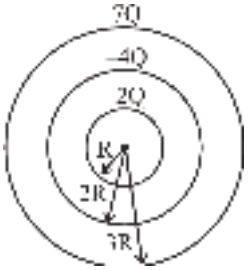
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is incorrect

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

8) Three concentric conducting shells of radii R, 2R & 3R having charges as shown. When inner most and outer most shells are connected by a thin wire without touching the middle shell, final



charges present on the inner and outer surfaces of the outer most shell will be :-

(1) +3Q, +4Q
(2) +4Q, +3Q
(3) +3Q, +5Q
(4) +5Q, +3Q

9)

A long solenoid having 200 turns per cm carries a current of 1.5 amp. A coil of 100 turns of cross-sectional area 3.14×10^{-4} m^2 is placed at the centre of solenoid having its axis parallel to the field produced by the solenoid. When the direction of current in the solenoid is reversed within 0.05 sec, the induced emf in the coil is :

(1) 0.48 V
(2) 0.048 V
(3) 0.0048 V
(4) 48 V

10) **Assertion (A)** :- For the purpose of calculation of moment of inertia, body's mass can be assumed to be concentrated at its centre of mass. **Reason (R)** :- Moment of inertia of rigid body about an axis passing through its centre of mass is zero.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(2) **(A)** is correct but **(R)** is not correct
(3) Both **(A)** & **(R)** incorrect
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

11) In a plane progressive harmonic wave particle speed is always less than the wave speed if- (λ = wavelength of wave)

(1) amplitude of wave is less than $\frac{\lambda}{2\pi}$
(2) amplitude of wave is greater than $\frac{\lambda}{2\pi}$
(3) amplitude of wave is less than λ

(4) amplitude of wave is greater than $\frac{\lambda}{\pi}$

12) A beam of white light is incident normally on a plane surface absorbing 70% of the light and reflecting the rest. If the incident beam carries 10W of power, then the force exerted by it on the surface will be :-

(1) 4.3×10^{-6} N

(2) 4.3×10^{-8} N

(3) 6.3×10^{-8} N

(4) 5.7×10^{-6} N

13) A liquid drop of diameter D breaks into 27 tiny drops (all 27 drops are identical). The resultant change in surface energy is –

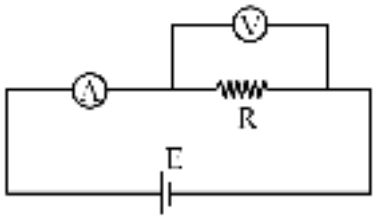
(1) $2\pi TD^2$

(2) $4\pi TD^2$

(3) πTD^2

(4) None of these

14) In a circuit, voltmeter and an ammeter are connected with a battery of emf E and external resistance R as shown in circuit. Ammeter resistance is $\frac{1}{10}$th of R and voltmeter resistance is 10 times that of R. Select the correct statement when compared with reading, when both voltmeter and ammeter are ideal.



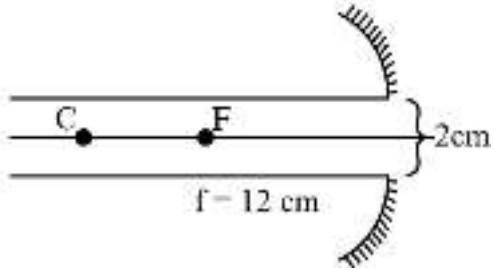
(1) Percentage error in reading of ammeter is 1%

(2) Percentage error in reading of voltmeter is 10%

(3) both (1) & (2) are correct.

(4) both (1) & (2) are wrong.

15) If a concave mirror having focal length 12 cm is divide in two parts as shown in diagram the separation between the images formed in upper and lower part will be :- (object is at C)

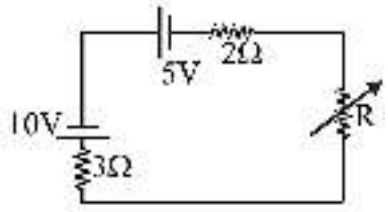


(1) 2 cm

(2) 4 cm

(3) 8 cm

(4) 6 cm

16) 

Find maximum power in resistance R :-

(1) 11.25 W

(2) 20 W

(3) 15.5 W

(4) 25 W

17) **Assertion :** Microwaves have less energy than visible waves.
Reason : Microwaves move faster than visible waves.

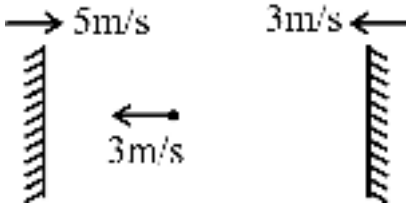
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

18) A particle collides elastically with two moving walls one by one as shown on a horizontal plane.



Find the final speed of the particle after two collisions :-

(1) 11 m/s

(2) 19 m/s

(3) 21 m/s

(4) 5 m/s

19) A satellite is launched in a circular orbit of radius R and another satellite is launched in circular orbit of radius 1.01 R. The time period of second satellite is different from that of the first satellite by :-

(1) 1.5 % Greater

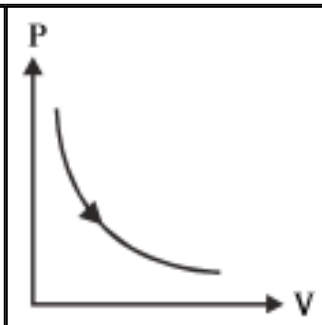
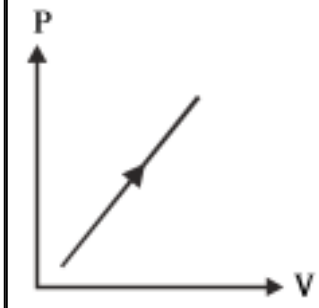
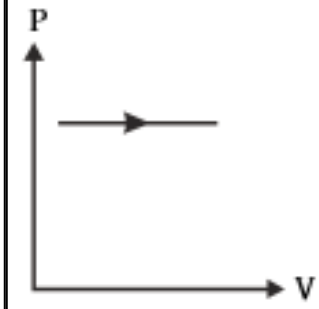
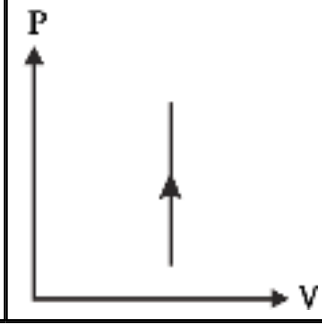
(2) 1% Lesser

(3) 1% Greater

(4) 1.5% Lesser

20)

Match the **correct** :-

(a)		(p)	Volume is constant
(b)		(q)	Volume is increasing
(c)		(r)	Temperature is constant
(d)		(s)	Pressure is constant

(1) a → q ; b → q, r ; c → q ; d → p

(2) a → r, p ; b → q, r ; c → q, s ; d → p, r

(3) a → q, r ; b → q ; c → q, s ; d → p

(4) None of these

21)

Energy released in the fission of a single $_{92}U^{235}$ nucleus is 200 MeV. The fission rate of $_{92}U^{235}$ fuelled reactor operating at a power level of 32 kW is :-

(1) 32×10^{13} /sec

(2) 10^{15} /sec

(3) 10^{11} /sec

(4) 10^{12} /sec

22) A glass sphere of density 2500 kg-m^{-3} is dropped in a liquid of density 1200 kg m^{-3}, having viscosity of 0.1 Pa-s. It attains a terminal velocity of 2.6 mm s^{-1}. Now same sphere when dropped in another liquid of density 1000 kg/m^3, of unknown viscosity η it attains a terminal velocity of 1.5 mm s^{-1}.
The value of η is :

(1) 0.15 Pa-s
(2) 0.20 Pa-s
(3) 0.24 Pa-s
(4) 0.36 Pa-s

23) If the surface of a metal succesively exposed to radiation of wavelength λ_1 = 350 nm and λ_2 = 450 nm, the maximum velocity of photoelectrons become half as compared to previous. The work function of this metal is :-

(1) 2.8×10^{-20} J
(2) 6.1×10^{-17} J
(3) 3.2×10^{-18} J
(4) 4.0×10^{-19} J

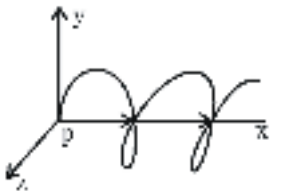
24) A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 (vertically). If the acceleration of the blocks is g/8, then the ratio of the masses is :

(1) 8 : 1
(2) 9 : 7
(3) 4 : 3
(4) 5 : 3

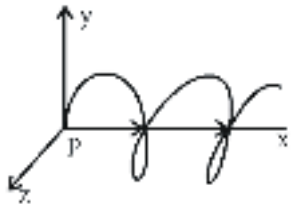
25) A travelling microscope is used to measure refractive index of a glass slab and following observations are obtained :
a. Cross - mark without slab : 2.150 cm
b. Cross - mark with slab : 3.350 cm
c. Top surface with lycopodium powder : 5.350 cm
The calculated value of refractive index of slab is :

(1) 1.500
(2) 1.550
(3) 1.600
(4) 1.650

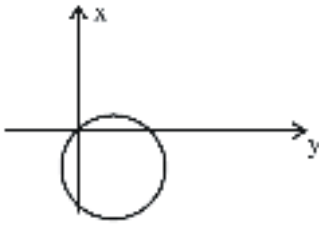
26) An electron is projected from origin with velocity $3.2\hat{i} + \hat{j}$ (m/s). A uniform magnetic field of $2\hat{k}$ tesla is present. The path followed by electron is correctly shown in ($m_e = 9 \times 10^{-31}$kg)

(1) 

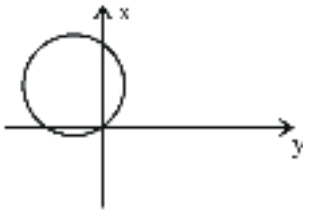
Helix on −ve side of z-axis

(2) 

Helix on -ve side of z-axis

(3) 

circle in xy plane

(4) 

circle in xy plane

27)

A balloon is moving upwards with constant velocity 10 ms^{-1}. It releases a stone which comes down to the ground in 11 s. The height of the balloon from the ground at the moment when the stone was dropped is (g = 10 m/s^2) :

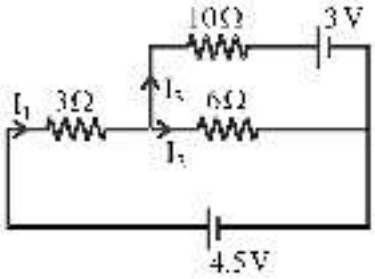
(1) 495 m
(2) 592 m
(3) 460 m
(4) 500 m

28) The orbital radius of electron in a H-atom changes from 2.12 Å to 4.77 Å. The H-atom has :-

(1) Absorbed a photon of energy 1.51 eV
(2) Absorbed a photon of energy 1.89 eV
(3) Emitted a photon of energy 10.2 eV
(4) Emitted a photon of energy 1.04 eV

29)

The currents I_1, I_2, & I_3 in the circuit are respectively (in A) :-



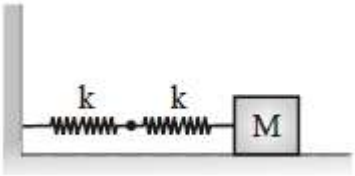
(1) $\frac{1}{2}, \frac{1}{2}, 1$

(2) 1, 1, 2

(3) $\frac{1}{2}, \frac{1}{2}, 3$

(4) $\frac{1}{2}, \frac{1}{2}, 0$

30) Two springs are connected to a block of mass M placed on a frictionless surface as shown below. If both the springs have a spring constant k, the frequency of oscillation of block is :



(1) $\frac{1}{2\pi}\sqrt{\frac{k}{M}}$

(2) $\frac{1}{2\pi}\sqrt{\frac{k}{2M}}$

(3) $\frac{1}{2\pi}\sqrt{\frac{2k}{M}}$

(4) $\frac{1}{2\pi}\sqrt{\frac{M}{k}}$

31) In a radio tuning RLC circuit, the half power frequencies are 100 MHz and 120 MHz. Find the quality factor.

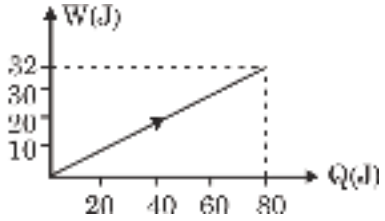
(1) 5.0

(2) 4.5

(3) 6.0

(4) 5.5

32) In the figure shown, amount of heat supplied to one mole of an ideal gas is plotted on the horizontal axis and amount of work done by gas is drawn on vertical axis. Assuming process be

isobaric, this gas can be :- 

(1) He

(2) CO

(3) CO_2

(4) NH_3

33) Two bodies of different materials and same surface area, coated by similar material, are heated to same temperature. They are allowed to cool down in same surroundings. If $\frac{dQ}{dt}$ represents rate of loss of heat and $\frac{d\theta}{dt}$ represents rate of fall of temperature, then for both

(1) $\frac{dQ}{dt}$ is same but $\frac{d\theta}{dt}$ is different

(2) $\frac{dQ}{dt}$ is different but $\frac{d\theta}{dt}$ is same

(3) $\frac{d\theta}{dt}$ as well as $\frac{dQ}{dt}$ is same

(4) $\frac{d\theta}{dt}$ as well as $\frac{dQ}{dt}$ is different.

34) **Assertion :-** Refractive index of turpentine oil with respect to air is more than that of water with respect to air.
Reason :- Mass density of turpentine oil is more than that of water.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

35) A polaroid examines two adjacent polarised light beams A and B of intensity I_A and I_B respectively, whose planes of polarisation are mutually at right angles. In one position of the polaroid, the beam B shows zero intensity. From this position, a rotation of 30° in polaroid shows the two beams of equal intensities. Find the intensity ratio I_A/I_B of the two beams.

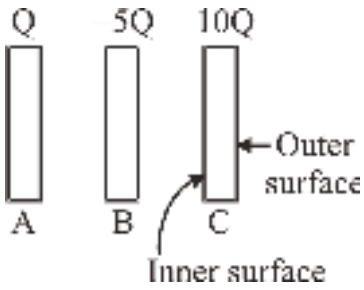
(1) 1 : 3

(2) $\sqrt{3} : 2$

(3) 1 : 1

(4) $1 : \sqrt{2}$

36) Three very large plates are given charges as shown in the figure. If the cross-sectional area of



each plate is same, the final charge distribution on plate C is :-

(1) +5Q on the inner surface, +5Q on the outer surface

(2) +6Q on the inner surface, +4Q on the outer surface

(3) +7Q on the inner surface, +3Q on the outer surface

(4) +8Q on the inner surface, +2Q on the outer surface

37) A student performs the experiment for determination of focal length of a concave mirror by using an optical bench by parallax method. The graph between u and v is obtained as shown in

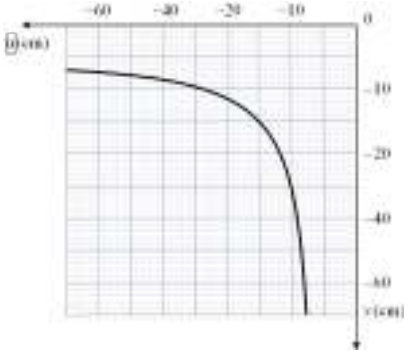


figure. The estimated value of focal length of concave mirror is :

(1) – 7.5 cm

(2) – 10 cm

(3) 12.5 cm

(4) – 15 cm

38) A particle suspended at the end of a spring executes simple harmonic motion with a period t_1, while the corresponding period for another spring is t_2. If the period of oscillation with the two springs in series is T, then :-

(1) $T = t_1 + t_2$

(2) $T^2 = t_1^2 + t_2^2$

(3) $T^{-1} = t_1^{-1} + t_2^{-1}$

(4) $T^{-2} = t_1^{-2} + t_2^{-2}$

39) What will be conductivity of pure silicon crystal at 300 K temperature. If electron hole pairs per cm^3 is 1.072×10^{10} at this temperature, $\mu_e = 1350\ cm^2$/volt-s and $\mu_h = 480\ cm^2$/volt-s :

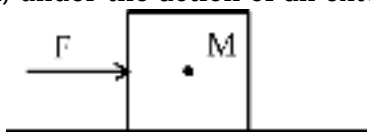
(1) 3.14×10^{-6} mho/cm

(2) 3×10^6 mho/cm

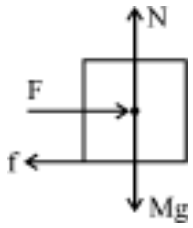
(3) 10^{-6} mho/cm

(4) 10^6 mho/cm

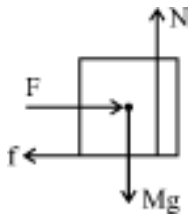
40) If the block shown in the figure is in equilibrium, under the action of an external force. Then choose the correct free body diagram of the block :-



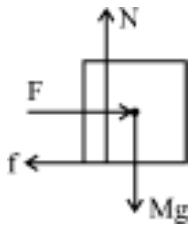
(1)



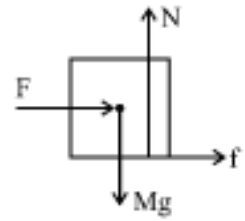
(2)



(3)



(4)



41) The area of hysteresis loop of a material is equivalent to 250 m^2. When 10 kg material is magnetised by an alternating field of 50 Hz then energy lost in one hour will be if the density of material is 7.5 gm/cm^3 :

(1) 6×10^4 J

(2) 6×10^4 erg

(3) 3×10^2 J

(4) 3×10^2 erg

42) A short bar magnet of magnetic moment 0.6 J T^{-1} is placed in a uniform magnetic field of 0.32 T. Magnet is in unstable equilibrium then the potential energy is :

(1) 0.192 J

(2) –0.192 J

(3) zero

(4) 0.096 J

43) If X = a + b, the maximum percentage error in the measurement of X will be :–

(1) $\left(\frac{\Delta a}{a}+\frac{\Delta b}{b}\right)\times 100\%$

(2) $\left(\frac{\Delta a}{a+b}-\frac{\Delta b}{a+b}\right)\times 100\%$

(3) $\left(\frac{\Delta a}{a+b}+\frac{\Delta b}{a+b}\right)\times 100\%$

(4) $\left(\frac{\Delta a}{a}\times\frac{\Delta b}{b}\right)\times 100\%$

44) The magnitude of electric field intensity at point (2, 0, 0) due to a dipole of dipole moment, $\vec{P}=\hat{i}+\sqrt{3}\hat{j}$ kept at origin is:
[where k = 1/4 π $\in_0$]

(1) $\frac{\sqrt{13}k}{8}$

(2) $\frac{\sqrt{13}k}{4}$

(3) $\frac{\sqrt{17}k}{8}$

(4) $\frac{\sqrt{7}k}{8}$

45) A tuning fork of frequency 300 Hz, when sounded over a resonance tube gives fundamental resonance when length of air column is 23 cm. The same tuning fork gives second overtone at 123 cm of air column. The end correction of tube is given by :

(1) 1.5 cm

(2) 2 cm

(3) 2.25 cm

(4) 2.5 cm

CHEMISTRY

1) The shortest wavelength of He^+ ion spectrum in Balmer series is x, then longest wavelength in the Paschen series of Li^{+2} ion spectrum is

(1) $\frac{36x}{5}$

(2) $\frac{16x}{7}$

(3) $\frac{9x}{5}$

(4) $\frac{5x}{9}$

2) 2 moles of A and 2 moles of B are mixed and the reaction is carried out at 400°C according to the equation $A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)}$, the equilibrium constant of the reaction is 4. Find the moles of C at

equilibrium :

(1) 0.5 mole
(2) 1 mole
(3) 2 mole
(4) 3 mole

3) Match the following column.

	Column-I (Salt undergoing salt-hydrolysis)		**Column-II (formula of pH)**
i	$HCOONH_4$	a	pH = 7
ii	NH_4NO_3	b	$pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$
iii	CH_3COOK	c	$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C$
iv	KCl	d	$pH = 7 - \frac{1}{2}pK_b - \frac{1}{2}\log C$

(1) i–b; ii–d; iii–a; iv–c
(2) i–b; ii–a; iii–c; iv–d
(3) i–d; ii–b; iii–c; iv–a
(4) i–b; ii–d; iii–c; iv-a

4) **Assertion (A) :-** If 30 ml of H_2 and 20 ml of O_2 react to form H_2O, 5 ml of H_2 is left at the end of the reaction.
Reason (R) :- In above reaction H_2 is limiting reagent.

(1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(2) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(3) Assertion is False but the Reason is True.
(4) Assertion is True but the Reason is False.

5) Which of the following is not a redox reaction :-

(1) $MnO_4^- + C_2O_4^{-2} \rightarrow Mn^{+2} + CO_2$
(2) $FeC_2O_4 \rightarrow Fe^{+3} + CO_2$
(3) $Zn + NO_3^- \rightarrow Zn^{+2} + NH_4^+$
(4) $VO + Fe_2O_3 \rightarrow V_2O_5 + FeO$

6) Standard enthalpy of vapourisation $\Delta_{vap}H°$ for water at 100°C is 40.66 kJ mol^{-1}. The change in internal energy of vaporization of water at 100°C (in kJ mol^{-1}) is :-

(1) +37.56
(2) –43.76

(3) +43.76

(4) +40.66

7) Consider the following reactions :

$C(s) + O_2(g) \rightarrow CO_2(g) + x\ kJ$

$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g) + y\ kJ$

The heat of formation of CO(g) is :

(1) –(x+y) kJ/mol

(2) (x–y) kJ/mol

(3) (y–x) kJ/mol

(4) (2y – x) kJ/mol

8)

Which indicator is used in the titration of sodium hydroxide against oxalic acid and what is the colour change at the end point ?

(1) Phenolphthalein, pink to yellow

(2) Alkaline $KMnO_4$, colourless to pink

(3) Phenolphthalein, colourless to pink

(4) Methyl orange, yellow to pinkish red colour

9) For a complex reaction $A \xrightarrow{k}$ products

E_{a_1} = 180 kJ/mol; E_{a_2} = 80 kJ/mol; E_{a_3} = 50 kJ/mol

Overall rate constant k is related to individual rate constant by the equation $k = \left(\frac{k_1.k_2}{k_3}\right)^{2/3}$.

Activation energy (kJ/mol) for the overall reaction is :-

(1) 100

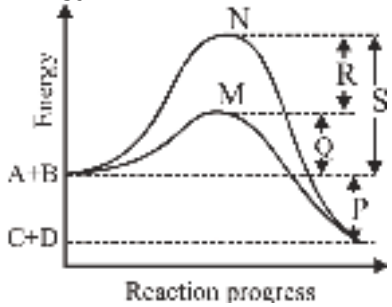
(2) 43.44

(3) 150

(4) 140

10) $A(g) + B(g) \rightleftharpoons C(g) + D(g)$

The curves M and N represent the variation of energy with reaction coordinate for the reaction in



presence and absence of catalyst respectively : Which value represents the activation energy (E_a) for the backward reaction in the absence of catalyst

(1) P

(2) Q

(3) S + P

(4) P + Q

11) The equivalent conductance of 1M benzoic acid is 12.8 Ω^{-1} cm^2 eq^{-1}. If the equivalent conductance of benzoate ion and H^+ ions at infinite dilution are 42 and 288.42 Ω^{-1} cm^2 eq^{-1} respectively. Its percentage of dissociation is :-

(1) 39%

(2) 0.38%

(3) 3.87%

(4) 8.27%

12) The e.m.f. of the following galvanic cells :-
(a) $Zn|Zn^{+2}(1M)||Cu^{2+}(1M)|Cu$
(b) $Zn|Zn^{2+}(0.1M)||Cu^{2+}(1M)|Cu$
(c) $Zn|Zn^{2+}(1M)||Cu^{2+}(0.1M)|Cu$
(d) $Zn|Zn^{2+}(0.1M)||Cu^{2+}(0.1M)|Cu$
are represented by E_1, E_2, E_3 and E_4 respectively. Which of the following is true?

(1) $E_1 > E_2 > E_3 > E_4$

(2) $E_3 > E_2 > E_1 > E_4$

(3) $E_3 > E_1 = E_4 > E_2$

(4) $E_2 > E_1 = E_4 > E_3$

13) 200 mL , $\frac{M}{10}$ H_2SO_4 is completely neutralized by decinormal NaOH solution, what will be the volume of NaOH required :-

(1) 200 mL

(2) 400 mL

(3) 100 mL

(4) 600 mL

14) What is molarity of 30 % w/w urea solution if density of solution is 1.2 g/mL :-
(M_w of urea = 60 g mol^{-1})

(1) 6 M

(2) 2 M

(3) 3 M

(4) 4 M

15) Total vapour pressure of mixture of 1mol A
(P^o_A = 150 torr) and 2 mol B (P^o_B = 300 torr) is 240 torr. In this case :-

(1) there is a positive deviation from Raoult's law.

(2) there is a negative deviation from Raoult's law.

(3) there is no deviation from Raoult's law.

(4) Can't predict

16) Which of the following is incorrect order for the indicated property?

(1) Cr < Mn < Fe: Ionisation energy

(2) O < Te < Se < S : Electron affinity

(3) C < N < F < O : 2nd Ionisation energy

(4) $O^- > S^-$: Ionisation energy

17)

List-I Elements (Electrons in K, L, M, N .... shells)		**List-II (Correct Statements)**	
(P)	W (2, 8, 6)	(1)	Paramagnetic with two unpaired electrons
(Q)	X (2,8,18,8)	(2)	f-block element
(R)	Y (2, 8, 18, 25, 8, 2)	(3)	10th group element
(S)	Z (2, 8, 18, 18)	(4)	Inert gas

(1) P-4; Q-1; R-2; S-3

(2) P-4; Q-1; R-3; S-2

(3) P-1; Q-4; R-2; S-3

(4) P-1; Q-4; R-3; S-2

18) **Statement (I) :** The dipole moment helps to predict whether a molecule is polar or non-polar.
Statement (II) : The dipole moment helps to predict the geometry of molecules.

(1) Statement I is correct and Statement II is incorrect

(2) Statement-I is incorrect and statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

19) How many of the following molecule/species does not exist ? Be_2, C_2^+, He_2^+, Ne_2, PH_5, BF_6^{3-}, SiF_6^{2-}

(1) 2

(2) 3

(3) 4

(4) 5

20) Which of the following energy term is not involved in Born-Haber cycle for NaCl formation ?

(1) Hydration energy of Na^+

(2) Second electron gain enthalpy of Cl

(3) Lattice energy of NaCl

(4) Both (1) & (2)

21) The number of lone pairs in XeF_2 are :-

(1) 3

(2) 6

(3) 9

(4) 12

22) A complex has a composition corresponding to the formula $CoBr_2Cl.4NH_3$. What is the actual formula if conductance measurements show two ions per formula unit? When excess silver nitrate is added, precipitate of AgCl is obtained but no precipitate of AgBr :-

(1) $[CoBrCl(NH_3)_4]Br$

(2) $[CoCl(NH_3)_4]Br_2$

(3) $[CoBr_2Cl(NH_3)_4]$

(4) $[CoBr_2(NH_3)_4]Cl$

23) For true statement use 'T' and for false use 'F'.
(a) $M(AB)_3$ pairs of enantiomers = 2
(b) Nature of bonding in $[Ni(CO)_4]$ can be properly (completely) explain by molecular orbital theory.
(c) Color of $KMnO_4$ is due to M → L charge transfer(MLCT)
(d) $[Co(NH_3)_6]Cl_3$ in this complex $[Co(NH_3)_6]^{+3}$ acts as lewis acid.

(1) T T F F

(2) T T T F

(3) T T T T

(4) F F F T

24) Which of the following has longest C – O bond length.

(1) $[V(CO)_6]^{+2}$

(2) $[V(CO)_6]^{-}$

(3) $[V(CO)_6]^{-2}$

(4) $[V(CO)_6]^{+1}$

25) Which of the following statements is/are correct.
(i) In 3d-series, all metals form "MO" types of ionic oxides except Sc.
(ii) Basic strength order of oxides of vanadium are $V_2O_3 > V_2O_4 > V_2O_5$.
(iii) CrO is basic while Cr_2O_3 is acidic in nature.
(iv) Mn (VI) shows disproportionation to Mn(VII) & Mn(IV) in acidic medium.

(1) (i) only

(2) (ii), (iii)

(3) (i), (ii)

(4) (i), (ii), (iv)

26) Name the gas that can readily decolourize acidified $KMnO_4$ solution :

(1) SO_2

(2) SO_3

(3) P_2O_5

(4) CO_2

27) Given below are two statements :
Statement-I : The chlorine oxides Cl_2O, ClO_2, Cl_2O_6 and Cl_2O_7 are highly reactive oxidising agents and tend to explode.
Statement-II : ClO_2 is used as bleaching agent for paper pulp and textiles and in water treatment.
Choose correct option.

(1) Statement-I correct, Statement-II incorrect

(2) Statement-I incorrect, Statement-II correct

(3) Both Statement-I and II are correct.

(4) Both Statement-I and II are incorrect

28) Choose the correct statement :-

(1) Naturally occurring carbon contains 2 stable isotopes, ^{12}C and ^{13}C.

(2) ^{14}C is a radioactive isotope with half life 5770 years.

(3) There are two isotopic forms of boron ^{10}B(19%) and ^{11}B (81%)

(4) All of the above

29) Match the List-I with List-II

	List-I (Cation)		**List-II (Group reagent)**
A.	Cu^{+2}	i.	$H_2S + HCl$
B.	Al^{+3}, Fe^{+3}	ii.	$(NH_4)_2CO_3 + NH_4OH$
C.	Co^{+2}, Ni^{+2}	iii.	$NH_4OH + NH_4Cl$
D.	Ba^{+2}, Ca^{+2}	iv.	$H_2S + NH_4OH$

Correct match is.

(1) A–iv; B–ii; C–iii; D–i

(2) A–i; B–iii; C–iv; D–ii

(3) A–iii; B–i; C–iv; D–ii

(4) A–i; B–iii; C–ii; D–iv

30) Which is incorrect statement ?

(1) Hg_2Cl_2 does not give chromyl chlorite test

(2) Na_2S salt give purple colour complex with sodium nitro prusside

(3) $CuSO_4$ give deep blue soluble complex with excess NH_4OH solution

(4) NH_3 gas give green colour ppt with Nesseler reagent

31) In the following sequence of reaction

$$CH_3CH_2OH \xrightarrow{P+I_2} A \xrightarrow[Ether]{Mg} B \xrightarrow{HCHO} C \xrightarrow{H_2O/H^+} D$$

the compound 'D' is :-

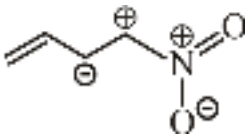
(1) Propanal

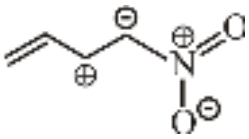
(2) Butanal

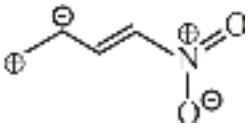
(3) n-butyl alcohol

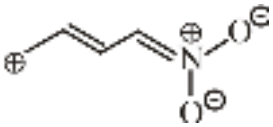
(4) n-propyl alcohol

32) Least stable resonating structure in the following :-

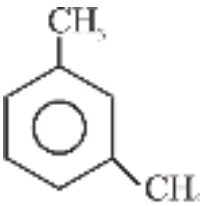
(1) 

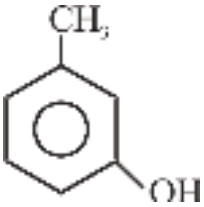
(2) 

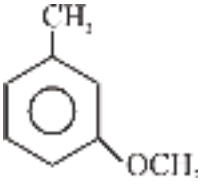
(3) 

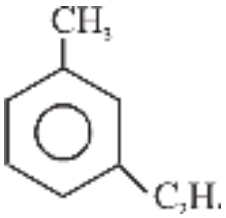
(4) 

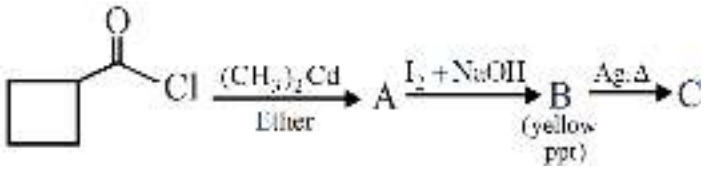
33) Which is most reactive for electrophilic substitution reaction :-

(1) 

(2) 

(3) 

(4) 

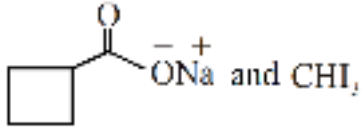
34) 

Identify B and C respectively ?

(1) CH_3I and $CH_3–CH_3$

(2) CHI_3 and $CH_3–CH_3$

(3) CHI_3 and $HC\equiv CH$

(4)  and CHI_3

35) Which of the following pair's give positive tollen's test :-

(1) Glucose, Sucrose

(2) Sucrose, Acetone

(3) Lactose, Fructose

(4) Acetophenone, Glucose

36) Identify x and y in the following reaction sequence

$C_2H_5Br \xrightarrow{x}$ product $\xrightarrow{y} C_3H_7NH_2$

(1) x = KCN , y = $LiAlH_4$

(2) x = KCN, y = H_3O^+

(3) x = CH_3Cl, y = $AlCl_3$/HCl

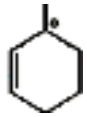
(4) x = $CH_3–NH_2$, y = HNO_2

37) Most stable free radical is :

(1) 

(2) 

(3) 

(4) 

38) S_N2 mechanism proceed with formation of transition state. Incorrect for transition state :-

(1) Carbon atom of transition state simultaneously bonded to incoming nucleophile and outgoing leaving group.

(2) It is highly unstable state and cannot be isolated.

(3) C-atom of transition state simultaneously bonded to five atom.

(4) C-atom of transition state is sp^3d hybridised.

39) How many total chiral compounds are possible on monochlorination of 2-methylbutane ?

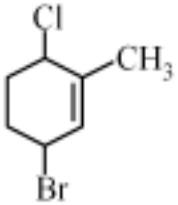
(1) 8

(2) 2

(3) 4

(4) 6

40) What is IUPAC name of the following compound.



(1) 3-Bromo-6-chloro-1-methyl cyclohexene

(2) 2-Bromo-5-chloro-6-methyl cyclohexene

(3) 6-Bromo-3-chloro-2-methyl cyclohexene

(4) 5-Bromo-2-chloro-1-methyl cyclohexene

41) Correct match is -

	Column-I		**Column-II**
(A)	Nitrogen of the organic compound is converted into $(NH_4)_2SO_4$	(p)	$Fe(SCN)_3$
(B)	Blue colour during Lassaigne's Test for Nitrogen is	(q)	$Fe_4[Fe(CN)_6]_3$
(C)	Compound formed in Lassaigne's test for N if S is also present	(r)	Kjeldahl's method
(D)	Compound responsible for violet colouration in Lassaigne's test for S.	(s)	$Na_4[Fe(CN)_5NOS]$

(1) A-r, B-s, C-q, D-p

(2) A-r, B-q, C-p, D-s

(3) A-p, B-s, C-q, D-r

(4) A-q, B-s, C-r, D-p

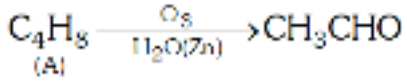
42) Which of the following do not show geometrical isomerism :-

(1) 1, 2-Dichloro- 1-pentene

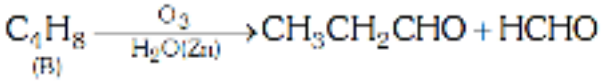
(2) 1, 3-Dichloro- 2-pentene

(3) 1,1-Dichloro- 1-pentene

(4) 1, 4-Dichloro- 2-pentene

43) 

(Only)



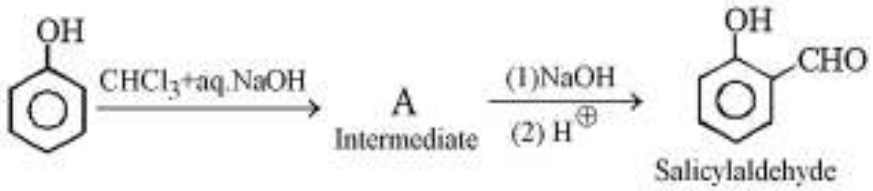
Correct stability order of A and B is–

(1) A > B

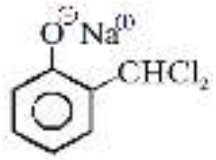
(2) A < B

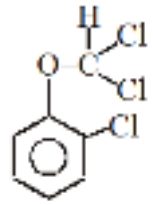
(3) A = B

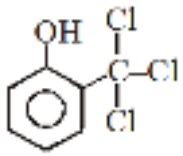
(4) Can't be predicted

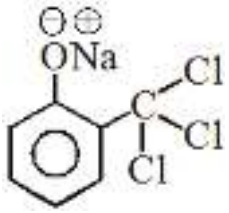


44) In following reaction : The most suitable structure of intermediate 'A' will be :

(1) 

(2) 

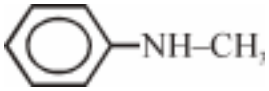
(3) 

(4) 

45) Which of the following can be prepared by Gabriel phthalimide synthesis :-

(1) CH_3–CH_2–NH_2

(2) 

(3) 

(4) $(CH_3)_2NH$

BIOLOGY

1) Match column - I & II and select the correct option :-

Column - I		Column - II	
(I)	Lion	(a)	*Felis*, a genus
(II)	Brinjal	(b)	*tigris*, specific epithet
(III)	Cat	(c)	*leo*, specific epithet
(IV)	Tiger	(d)	*Solanum,* a genus

(1) I – b, II – a, III – d, IV – c

(2) I – c, II – d, III – a, IV – b

(3) I – c, II – a, III – d, IV – b

(4) I – b, II – d, III – a, IV – c

2) Read the following statements(A-D) carefully-
(A) For plants, scientific names are based on agreed principles and critaria, which are provided in ICBN.
(B) The scientific names ensure that each organism has only one name.
(C) Biologist follow universally accepted principles to provide scientific names to known organisms.
(D) Binomial nomenclature was given by Linnaeus.
How many of the above statements are correct ?

(1) One

(2) Two

(3) Four

(4) Three

3) **Statement-I :** Archaebacteria differ from other bacteria in having different cell wall structure and this feature is responsible for their survival in extreme conditions.
Statement-II : Heterotrophic bacteria are most abundant in nature. The majority are important decomposers.

(1) Only Statement-I is incorrect

(2) Both statement-I and statement-II are correct

(3) Only Statement-II is incorrect

(4) Both statement-I and Statement-II are incorrect.

4) Study the following statements :

A. The fungi constitute a unique kingdom of heterotrophic organisms.
B. Fungi show a great diversity in morphology and habitat.
C. Most of the deuteromycetes are decomposer of litter and help in mineral cycle.
D. The deuteromycetes reproduce only by asexual spores known as ascospore.
Select the correct option :-

(1) All are correct except 'D'

(2) All are incorrect except 'B'

(3) Only A and B are correct

(4) Only A and D are incorrect

5) Which of the following groups of algae belongs to class Rhodophyceae ?

(1) *Laminaria*, *Chlorella*, *Volvox*, *Fucus*

(2) *Gelidium*, *Porphyra*, *Dictyota*, *Volvox*

(3) *Polysiphonia*, *Gelidium*, *Porphyra*, *Gracilaria*

(4) *Sargassum*, *Fucus*, *Chara*, *Volvox*

6)

Read the following statements :
Statement A: Bryophytes live in soil but depend on water for sexual reproduction.
Statement B: The male gametes (sperm) of bryophytes are biflagellated.

Choose the correct option :-
(1) Both the statements A and B are incorrect

(2) Both the statements A and B are correct

(3) The statement A is correct but B is incorrect

(4) The statement A is incorrect but B is correct

7) The leaves of gymnosperms are well-adapted to withstand extremes of temperature, humidity and wind. In conifers, what are the xerophytic characters?

(1) Needle-like leaves

(2) Thick cuticle

(3) Sunken stomata

(4) All of these

8) Read the following statements
Statement -I : Mango and coconut fruit are developed from monocarpellary gynoecium with superior ovary and are one seeded.
Statement -II : The embryo is made up of an embryonal axis with a radicle and a plumule and one cotyledon in maize.
Choose the correct answer from the option given below :

(1) Both Statement-I and II are incorrect

(2) Statement-I is correct but Statement-II is incorrect

(3) Statement-I is incorrect but Statement-II is correct

(4) Both Statement-I and II are correct

9) Which of the following statement is wrong for dorsiventral leaf ?

(1) Spongy tissue is situated towards abaxial surface

(2) Cuticle on upper surface is thick

(3) Palisade tissue is situated towards abaxial surface

(4) Present at right angle to stem

10) Arrange the following cell layers in dicot stem from outer to inner side :
(a) Hypodermis (b) Pericycle
(c) Endodermis (d) Epidermis
(e) Pith

(1) (d), (a), (c), (e), (b)

(2) (e), (c), (b), (a), (d)

(3) (b), (a), (d), (c), (e)

(4) (d), (a), (c), (b), (e)

11)

Read the following statements :-
(a) The quasifluid nature of protein enable lateral movement of lipid within the overall bilayer.
(b) Primary wall is capable of growth.
(c) Plasma membrane is symmetrical structure.
(d) E.R. divide the intracellular space into two distinct compartment.
(e) Lysosome enzymes are activated at pH 7.2.

How many statements are **not** correct ?

(1) 2

(2) 1

(3) 3

(4) 4

12) From the column-I, column-II, column-III how many of the following are correctly match –

	Column-I	**Column-II**	**Column-III**
(A)	Ribosome	Composed of r-RNA and protein	Site of protein synthesis
(B)	ER	Network of tiny tubular structure scattered in the cytoplasm	Site for synthesis for lipid
(C)	Kinetochore	Disc shape structure present on secondary constriction of chromosome	Site of attachment for spindle fibres

(D)	Microbody	Membrane bound minute vesicle	Present only in plant cell

(1) 3

(2) 2

(3) 1

(4) 4

13)

Choose the incorrect statements :-
(A) Cells in G_0 stage are metabolically inactive.
(B) Meiosis involves two cycle of DNA replication
(C) Mitosis restores the nucleo–cytoplasmic ratio of cells
(D) Duration of cell cycle may vary from organism to organism and also from cell to cell.

Options :-

(1) A,C

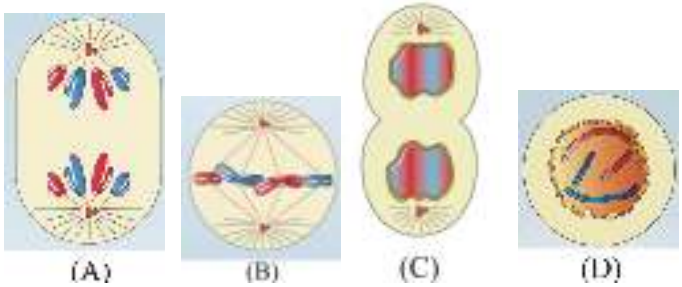
(2) C,D

(3) B,C

(4) A,B

14)

The four figures (A, B, C and D) given below represent four different stages of mitotic cell division. Which one of these is correctly identified in the options given, alongwith its correct description and key events ?



		Stage	Description	Key event
(1)	D	Prophase	Followed by G_2 Phase	Bivalent Chromosomes align on the equatorial plate
(2)	B	Metaphase	Condensation of chromosomes completed	Daughter chromosomes move to opposite poles
(3)	C	Telophase	Longer and highly complex	Homologous chromosomes separate

(4)	A	Anaphase	Centromeres are towards poles and arms trailing behind	Centromere split and chromatids separate

(1) 1

(2) 2

(3) 3

(4) 4

15) Match the following & choose correct option :-

a.	Metaphase-I	(i)	Splitting of centromere of each chromosome
b.	Anaphase-I	(ii)	Separation of homologous chromosome
c.	Telophase-I	(iii)	Alignment of bivalents on equatorial plate
d.	Anaphase-II	(iv)	Reformation of nuclear membrane

(1) a-iii, b-ii, c-iv, d-i

(2) a-iii, b-i, c-iv, d-ii

(3) a-iv, b-iii, c-ii, d-i

(4) a-i, b-ii, c-iii, d-iv

16) **Assertion :** Isomerases catalysing inter-conversion of optical geometric or positional isomers.
Reason : Almost all enzymes are proteins.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

17) An amino acid under certain condition have both positive & negative charge simultaneously in the same molecule such a form of amino acid is called?

(1) Acidic form

(2) Zwitter ionic form

(3) Basic form

(4) Aromatic form

18)

Find the wrong match pair :-

(1)	Secondary metabolite	Anthocyanin
(2)	Chitin	Polysaccharide
(3)	Protein	Insulin
(4)	Cellulose	Hetero polysaccharide

(1) 1

(2) 2

(3) 3

(4) 4

19) Given below are two statement :-
Statement I :- Two molecules of NADH are produced from the metabolism of two molecules of pyruvic acid during link reaction.
Statement II :- Acetyl CoA enters a cyclic pathway, more commonly called as kreb's cycle which scientifically elucidated by Hans Kreb at first.

(1) Statement I and II both are correct.

(2) Statement I and II both are incorrect.

(3) Only statement I is correct.

(4) Only statement II is correct.

20) TCA cycle starts with which type of reaction:-

(1) Dehydrogenation

(2) Decarboxylation

(3) Condensation

(4) Oxidation

21) Match the column-I, II & III and choose the option showing correct match :-

	Column-I PGR		**Column-II Chemical Nature**		**Column-III Physiological effect**
(I)	Cytokinin	(a)	Terpenes	(i)	Speed up malting process
(II)	ABA	(b)	Indole Compounds	(ii)	Initiate rooting in stem cutting
(III)	Auxin	(c)	Carotenoid derivatives	(iii)	Induce seed dormancy
(IV)	Gibberellin	(d)	Adenine derivatives	(iv)	Delay leaf senescence

Options :-

(1) I-d-iv, II -b-ii, III-c-iii, IV-a-i

(2) I-d-i, II -c-ii, III-b-iii, IV-a-iv

(3) I-a-iv, II-b-iii, III-c-ii, IV-d-i

(4) I-d-iv, II-c-iii, III-b-ii, IV-a-i

22)

Which of the following is not true regarding plasticity in plants ?

(1) In cotton, leaves of juvenile plant is similar to mature plant

(2) Heterophylly in butter cup is an example of plasticity

(3) Plants show plasticity in response to environment

(4) Cotton, coriander, larkspur and butter cup show plasticity

23) Select the **correct** statements ?
(A) F_0 is peripheral membrane protein complex that forms the channel through which protons cross the inner membrane of Mitochondria.
(B) The F_1 headpiece is an integral membrane protein complex and contains the site for synthesis of ATP from ADP and inorganic phosphate in Mitochondria.
(C) CF_0 is embedded in the thylakoid membrane and forms a transmembrane channel in Chloroplast.
(D) CF_1 protrudes on the inner surface of the thylakoid membrane on the side that faces the lumen.

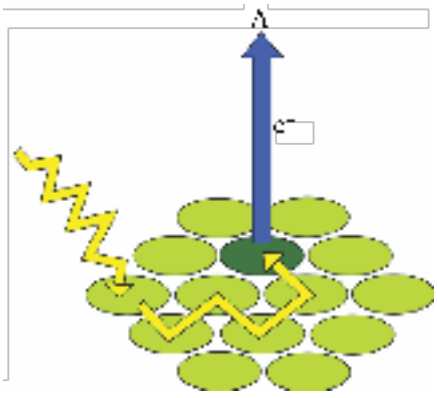
(1) (A) , (B) , (C), (D) all are correct.

(2) (C) only

(3) (A), (B) and (C) only

(4) (C) and (D) only

24) Consider the below figure and identify 'A'



(1) Reaction center

(2) Chl-a

(3) Primary e^- acceptor

(4) ETS

25) Name the scientist who pointed out the importance of different wavelengths of light using a green algae and aerobic bacteria :–

(1) Priestley

(2) Ingen-Housz

(3) K.V. Thimann

(4) Englemann

26) In Angiosperm, if haploid number of chromosome is 12 then what will be the no. of chromosomes in integuments and synergids ?

(1) 12, 12

(2) 24, 12

(3) 24, 24

(4) 12, 24

27) Which of the following event is not included under post pollination events ?

(1) Triple fusion

(2) Development of endosperm

(3) Transformation of ovary into fruit

(4) Formation of embryo sac

28) Due to efforts of Ramdev Misra, government of India established the NCEPC (1972) which stands for -

(1) National committee for environmental protection and coordination.

(2) National committee for environmental planning and coordination.

(3) National committee for environmental protection and conservation.

(4) National committee for environmental planning and conservation.

29) Which of the following is not a possible outcome of competition ?

(1) Competitive exclusion

(2) Co existence through niche partitioning

(3) Competitive release

(4) Complete niche overlap with stable co-existence

30) Which are **correct** from following ?
(A) Population ecology links ecology to population genetics and evolution.
(B) Birth rate and death rate are features of an organism.
(C) If a new habitat is just being colonised, immigration may contribute more significantly to population growth than birth rate.
(D) The exponential growth model is considered a more realistic one.

(1) A & B are correct

(2) B & C are correct

(3) C & D are correct

(4) A and C are correct

31)

	Column-I		Column-II		Column-III
(1)	Producer	(i)	Second trophic level	(a)	Lion
(2)	Primary consumer	(ii)	Third trophic level	(b)	Phytoplankton
(3)	Secondary consumer	(iii)	Fourth trophic level	(c)	Birds
(4)	Tertiary consumer	(iv)	First trophic level	(d)	Zooplankton

(1) 1-iii-b, 2-i-c, 3-iv-a, 4-ii-d

(2) 1-iii-a, 2-ii-b, 3-iv-d, 4-i-c

(3) 1-iv-a, 2-ii-d, 3-iii-b, 4-i-c

(4) 1-iv-b, 2-i-d, 3-ii-c, 4-iii-a

32) Which of the following statements is incorrect ?

(1) David Tilman's experiments support role of diversity in ecosystem stability

(2) Paul Ehrlich compared ecosystem stability to rivets on an airplane

(3) In Tilman's study, low diversity plots showed greater biomass stability

(4) Rivet popper hypothesis was proposed by Paul Enrlich.

33) Match the list-I with list-II

	List-I		List-II
(a)	Robert May	(i)	species area relationship
(b)	Alexander von Humboldt	(ii)	long term ecosystem experiment using out door plots
(c)	Paul Ehrlich	(iii)	global species diversity at about 7 Million
(d)	David Tilman	(iv)	Rivet popper hypothesis

Choose the correct answer from the option given below

(1) a-ii, b-iii, c-i, d-iv

(2) a-iii, b-i, c-iv, d-ii

(3) a-i, b-iii, c-ii, d-iv

(4) a-iii, b-iv, c-ii, d-i

34) Given below are two statements :
Statement I :- Amazon rain forest are called the lungs of the Earth.
Statement II :- Over exploitation of natural resources is the most important cause of biodiversity loss.
In the light of above statements, choose the correct answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

35) In which of the given process, both of the given strand of nucleic acid act as template

(1) Transcription

(2) Translation

(3) DNA Replication

(4) Reverse transcription

36) Which one of the following pairs of nitrogen bases of nucleic acid, is wrongly matched with the category mentioned

(1) Guanine, Adenine — Purines

(2) Adenine, Thymine — Purines

(3) Thymine, Uracil — Pyrimidines

(4) Uracil, Cytosine — Pyrimidines

37) **Assertion (A) :-** Regulation of lac operon by repressor is refered to to as negative regulation.
Reason (R) :- The repressor protein binds to the operator region of the operon and prevents RNA polymerase from transcribing the operon.

(1) Assertion is true but reason is false

(2) Both A & R are false

(3) A & R both are true but R is not correct explanation of A

(4) A & R both are true and R is correct explanation of A

38)

	Column-I		**Column-II**
(A)	Antisense strand	(i)	Synthesize hn-RNA
(B)	Ribonucleoside triphosphate	(ii)	Termination factor
(C)	Rho (ρ) factor	(iii)	Template for transcription
(D)	RNA polymerase-II	(iv)	Substrate for RNA synthesis

(1) A-iii, B-ii, C-iv, D-i

(2) A-iii, B-i, C-ii, D-iv

(3) A-iii, B-iv, C-ii, D-i

(4) A-i, B-ii, C-iii, D-iv

39) James Watson and Francis Crick, based on the X-ray diffraction data produced by Wilkins and Rosalind Franklin, proposed a very simple but famous Double Helix model for the structure of DNA.
Statement-I :- One of the hallmarks of their proposition was base pairing between the two strands of polynucleotide chains.
Statement-II :- This proposition was also based on the observation of Erwin Chargaff that for a

double stranded DNA, the ratios between Adenine and Thymine and Guanine and Cytosine are constant and equals one.

(1) Both **Statement I** and **Statement II** are correct.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct.
(4) Both **Statement I** and **Statement II** are incorrect.

40) If the ratio (A + G)/(T + C) in one strand DNA is 0.7 what is the same ratio in complementary strand ?

(1) 0·7
(2) 1·43
(3) 0·35
(4) 1

41) Human male can be heterozygous for -

(1) Colourblindness
(2) Haemophilia
(3) Thalassemia
(4) Down's syndrome

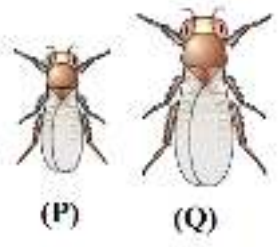
42) Modified allele is equivalent to unmodified allele if it produces -

(1) Normal enzyme
(2) Non-functional enzyme
(3) No enzyme at all
(4) No RNA at all

43) Find out correct statement :
(a) Contrasting traits show blending at F_1 stage.
(b) In honey bee males produce sperm by meiosis
(c) In birds male heterogamy is found
(d) Phenylketonuria is an example of pleiotrophy

(1) a, b, c
(2) Only a, b
(3) c and d
(4) Only d

44) Fully grown male and female drosophila are shown in figures (P & Q). Select the correct



statement, regarding this

(1) P is male Drosophila with sex chromosome XY.

(2) P is female Drosophila with sex chromosome XX.

(3) Q is male Drosophila with sex chromosome ZW.

(4) Q is female Drosophila with sex chromosome ZZ.

45) In the F_2 generation of a dihybrid cross, what will be the ratio between plants having new phenotypes and new genotypes-

(1) 5 : 3

(2) 3 : 7

(3) 5 : 1

(4) 1 : 7

46) The presence of characteristic flagellated choanocytes and a cellular level of organization are features of which phylum?

(1) Coelenterata

(2) Porifera

(3) Ctenophora

(4) Platyhelminthes

47) Mouth is mostly terminal have four pairs of gills which are covered by an operculum is not a character of :

(1) *Scoliodon*

(2) *Exocoetus*

(3) *Hippocampus*

(4) *Labeo*

48) Which of the following statements is incorrect regarding platyhelminthes ?

(1) They are mostly endoparasites found in animals.

(2) Flame cells help in osmoregulation and excretion.

(3) They are acoelomates and bilaterally symmetrical.

(4) They possess a well developed circulatory system.

49) Match the column-I with column-II and select correct answer about them

	Column-I		**Column-II**

a	*Sycon*	i.	Pseudocoelomate
b	*Obelia*	ii.	3-pairs of legs
c	*Ascaris*	iii.	Radula
d	*Locusta*	iv.	Cellular level of organisation
e	*Pila*	v.	Metagenesis

(1) a – i, b – ii, c– iii, d–iv, e–v

(2) a – iv, b – v, c– iii, d–i, e–ii

(3) a – iv, b – v, c– i, d–ii, e–iii

(4) a – v, b – iv, c– ii, d–iii, e–i

50) **Statement-I :-** The most distinctive feature of echinoderms is the presence of water transport system which helps in locomotion, capture and transport of food and respiration
Statement-II :- In molluscs the body is covered by a calcareous shell and segmented with a distinct head, muscular foot and viseral hump.

(1) Both statement-I and II are correct.

(2) Both statement-I and II are incorrect.

(3) Only statement-I is correct.

(4) Only statement-II is correct.

51)

Only tissue which originated from all these germininal layers' ectoderm, mesoderm and endoderm is :-

(1) Connective tissue

(2) Epithelium tissue

(3) Muscular tissue

(4) Neural Tissue

52) Find out incorrect statement about Cockroach.

(1) Three ganglia lies in thorax, six in abdomen

(2) Exchange of gases takes place at tracheloes

(3) Mosaic vision with more resolution but less sensitivity

(4) Supra-oesophageal ganglion supplies nerve to antennae & compound eye

53) In cockroach digestive juice is secreted by :-

(1) Proventriculus

(2) Hepatic Caeca

(3) Malpighian Tubule

(4) gizzard

54) Which of the following is sexual dimorphic character of frog.

(1) Salivary gland

(2) Tympanum

(3) Vocal sacs

(4) Nictitating membrane

55) **Assertion (A) :** The duodenum of frog receive bile from gall-bladder and pancreatic juice from pancreas through a common bile duct.
Reason (R) : Bile emulsifies fat and pancreatic juice digest carbohydrates and proteins.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Both Assertion & Reason are incorrect.

(4) Assertion is correct but the Reason is incorrect.

56) **Statement-I :** Plasma is a straw coloured, viscous fluid constituting nearly 45 percent of the blood.
Statement-II : Neutrophils and monocytes are phagocytic cells.

(1) Both Statements are correct.

(2) Both Statements are incorrect.

(3) Statement-I is correct while Statement-II is incorrect.

(4) Statement-I is incorrect while Statement-II is correct.

57) When the heart muscle is suddenly damaged by an inadquate blood supply, this condition called as -

(1) Heart failure

(2) Heart attack

(3) Cardiac arrest

(4) CAD

58) Read column-I & column-II and identify their correct match :-

Column-I		**Column-II**	
(A)	Persons with 'O' blood group	(i)	Anti-B antibodies in plasma
(B)	Persons with 'A' blood group	(ii)	Anti-A antibodies in plasma
(C)	Persons with 'B' blood group	(iii)	A, B antigens on RBCs
(D)	Persons with 'AB' blood group	(iv)	Anti-A, Anti-B antibodies in plasma
		(v)	Universal donors
		(vi)	Universal recipients

(1) (A)-(iv)(v), (B)-(i), (C)-(ii), (D)-(iii)(vi)

(2) (A)-(iii)(vi), (B)-(ii), (C)-(i), (D)-(iv)(v)

(3) (A)-(iv)(v), (B)-(ii), (C)-(i), (D)-(iii)(vi)

(4) (A)-(iii)(vi), (B)-(i), (C)-(ii), (D)-(iv)(v)

59) Which one do not contract during normal inhalation ?

(1) Diaphragm and external intercostal muscle.

(2) Diaphragm

(3) External intercostal muscle

(4) Internal intercostal muscle.

60) Read the following assertion and reason carefully and answer accordingly.
Assertion : A gradient is present for CO_2 is from tissues to blood and blood to alveoli.
Reason : As the solubility of CO_2 is 20-25 times lower than that of O_2, the amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much lower compared to that of O_2.

(1) Both assertion and reason are correct and reason is correct explanation of assertion.

(2) Assertion and reason both are correct, reason is not correct explanation of assertion.

(3) Assertion is incorrect, reason is correct.

(4) Assertion is correct, reason is incorrect.

61) Match the column-I with column-II

	Column-I		**Column-II**
(a)	Vital capacity	(i)	ERV + RV
(b)	Functional residual capacity	(ii)	TV + IRV + ERV
(c)	Inspiratory capacity	(iii)	TV + IRV
(d)	Total lung capacity	(iv)	VC + RV

Choose correct option :-

(1) a-ii, b-iii, c-i, d-iv

(2) a-iii, b-ii, c-i, d-iv

(3) a-ii, b-i, c-iii, d-iv

(4) a-i, b-ii, c-iv, d-iii

62) Cerebral hemispheres are connected by a tract of nerve fibres called as -

(1) Corpus albicans

(2) Thalamus

(3) Corpus callosum

(4) Cerebellar peduncle

63) Read the following statements and choose the correct option -

(1) Chemical synapse are faster than electrical.
(2) Electrical synapse are rare in our body.
(3) Rise in the stimulus induced permeability to K^+ is extremely long lived.
(4) Resting potential is also known as nerve impulse.

64) Blood glucose level is not increase by which hormone ?

(1) Adrenaline
(2) Glucagon
(3) Growth Hormone
(4) Calcitonin

65) Pair of hormone which interact with membrane bound receptors :-

(1) Cortisole & Estradiol
(2) Testosterone & Progesterone
(3) Glucagon & Thyroid Hormone
(4) Insulin & Epinephrine

66) Read the following statements -
(A) Hormones are non-nutrient chemicals which act as intercellular messengers and are produced in large amounts.
(B) Adrenal, pancreases, parathyroid and thymus are the organised exocrine bodies in our body.
(C) Invertebrates possess very simple endocrine systems with few hormones.
(D) Endocrine glands lack ducts and are hence, called ductless glands.
How many of the above statements are correct?

(1) Three
(2) Four
(3) Only One
(4) Two

67) Read the following Assertion (A) and Reason (R) and choose the correct option.
Assertion (A) : The pineal gland is located on the dorsal side of forebrain.
Reason (R) : Pineal secretes a hormone called melatonin.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is incorrect.
(4) Both (A) and (R) are incorrect.

68) Which type of joint occurs between carpal and metacarpal of thumb ?

(1) Hinge joint
(2) Pivot joint

(3) Glinding joint

(4) Saddle joint

69) Rapid spasms in museles due to low Ca^{++} in body fluid know as.

(1) Tetany

(2) Muscular dystrophy

(3) Myasthenia gravis

(4) Arthritis

70) Which hormones level reaches peak during ovulatory phase of menstrual cycle ?

(1) Progesterone

(2) HCG

(3) Relaxin

(4) LH

71) Ovulation in human a female takes place on 17th day of first menstrual bleeding. The duration of menstrual cycle in that female should be of:

(1) 05 Days

(2) 19 Days

(3) 28 Days

(4) 31 Days

72) Match the following columns-

Column-I		**Column-II**	
a.	Prolactin	(i)	Functioning of Mammary gland
b.	Estrogen	(ii)	Milk ejection
c.	Oxytocin	(iii)	Breast development
d.	Progesterone	(iv)	Milk synthesis

(1) a-(i), b-(iii), c-(iv), d-(ii)

(2) a-(ii), b-(iii), c-(iv), d-(i)

(3) a-(ii), b-(iv), c-(i), d-(iii)

(4) a-(iv), b-(iii), c-(ii), d-(i)

73) **Statement-1:** The clitoris is a tiny finger-like structure which lies at the upper junction of the two labia minora above the urethral opening.
Statement-2: A pair of ovaries along with a oviduct, uterus, cervix, vagina and the external genitalia located in pelvic region are integrated structurally and functionally to support the processes of ovulation, fertilisation, pregnancy, birth and child care.

(1) Both statement-1 and statement-2 are correct.

(2) Statement-1 is correct but statement-2 is incorrect.

(3) Both statement-1 and statement-2 are incorrect.

(4) Statement-1 is incorrect but statement-2 is correct.

74) Which of the following is a non-medicated IUD?

(1) Lippes loop

(2) Multiload 375

(3) Progestasert

(4) LNG-20

75) Psychological state of person due to consumption of certain drugs, which cause euphoria, temporary feeling of well being etc. is known as.

(1) Addiction

(2) Dependence

(3) Depression

(4) Isolation

76) Pneumonia, AIDS, Cancer, Typhoid, T.B., Malaria, Common Cold, Polio, Diphtheria, Plague. How many of the given example are bacterial disease ?

(1) Six

(2) Four

(3) Five

(4) Seven

77) In which type of disease body loss the ability to differentiate between self and non self cells ?

(1) Cancer

(2) AIDS

(3) Rheumatoid arthritis

(4) Malaria

78) **Statement-I :** Passive immunity provides immediate protection to an individual but it is short lived and does not produce immunological memory against future infections.
Statement-II : Active immunity develops more slowly but is long lasting because the body generates both plasma cells producing antibodies and memory cells. Capable of rapid response.

(1) Both statement-I and statement-II are true

(2) Both statement-I and statement-II are false

(3) Only statement-I is true

(4) Only statement-II is true

79) Malpighian tubules are the excretory structure of :-

(1) Prawn

(2) Cockroach

(3) Earthworm

(4) *Planaria*

80) Maximum and Minimum reabsorption occurs respectively in :-

(1) PCT and DCT

(2) DCT and PCT

(3) PCT and Ascending limb of loop of Henle

(4) LOH and PCT

81) Read the following statement (A to D) :-
(A) Ammonia is the least toxic and requires least amount of water for its elimination
(B) Ammonia is the most toxic and requires maximum amount of water for its elimination
(C) Uric acid is least toxic and require least amount of water for its elimination.
(D) Uric acid is most toxic and require maximum amount of water for its elimination.
Choose the correct answer :-

(1) A & C correct

(2) B & D correct

(3) A & D correct

(4) B & C correct

82) The phenomenon of the evolution of different species in a given geographical area starting from a point and spreading to other habitats is called :-

(1) Saltation

(2) Co-evolution

(3) Natural selection

(4) Adaptive radiation

83) Identify the correct match from column-I and column-II

	Column-I		**Column-II**
A.	Invertebrates formed	(i)	350 mya
B.	Sea weed plants existed	(ii)	65 mya
C.	Jawless fish evolved	(iii)	500 mya
D.	Extinction of Dinosaurs	(iv)	320 mya

(1) A-iv, B-i, C-iii, D-ii

(2) A-iv, B-iii, C-i, D-ii

(3) A-iii, B-iv, C-i,D-ii

(4) A-i, B-iii, C-ii, D-iv

84) **Assertion :** hCG and hPL are produced in women only during pregnancy.
Reason : These hormones are produced by placenta.

(1) Both assertion and reason are true and the reason is a correct explanation of the assertion.

(2) Both assertion and reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion and reason are false

85) Match the following columns

Column-I (Biocontrol agent)		**Column-II (pests)**	
A.	Ladybird beetle	(i)	mosquito larvae
B.	Dragonflies	(ii)	aphids
C.	*Baculovirus*	(iii)	arthropods
D.	*Trichoderma sp.*	(iv)	plant pathogen

(1) A-i,B-ii,C-iv,D-iii

(2) A-i,B-iii,C-iv,D-ii

(3) A-ii,B-i,C-iii,D-iv

(4) A-ii,B-i,C-iv,D-iii

86) With regard to insulin choose correct statements:-
(a) C-peptide is not present in mature insulin.
(b) The insulin produced by r-DNA technology has C-peptide
(c) The Pro Insulin has C-peptide.
(d) A peptide and B peptide of insulin are inter-connected by disulphide bridges.
Choose the correct answer from the options given below :-

(1) b and d only

(2) b and c only

(3) a, c and d only

(4) a and d only

87) Competent host cell is required for transformation with rDNA. Following processes are used to make the host cell competent except :-

(1) Treating them with specific concentration of divalent ion like Ca^{++}

(2) By incubating the cells with rDNA on ice, followed by placing them briefly at 42ºC & then putting back on ice

(3) Agarose gel electrophoresis

(4) Both (1) & (2)

88) Which of the following is correct with respect to the diagram given below ?



(1) Root of Vitamin 'A' enriched rice

(2) Root of plant that is more tolerant to abiotic strsses (cold, drought, salt, heat).

(3) Roots of a typical control plant infected by nematede

(4) Transgenic plant roots 5 days after deliberate infection of nematoda but protected through novel mechanism

89) Which of the following is not the use of transgenic animal ?

(1) Study of normal physiology and development

(2) Testing of vaccine safety

(3) Production of biological products

(4) Molecular diagnosis

90) Fill up the blanks.
I. Using *Agrobacterium* vectors, ...A.... specific genes are introduced into the host plant.
II. RNAi takes place in all eukaryotic organisms as a method of .....B.....
III. The genes .....C.... and .....D.... control cotton bollworms.
IV. ....E... has been introduced in Bt corn to protect the same corn borer.
A to E in above statements can be:-

(1) A-fungal, B-pest resistance,
C-cry II Ab, D-cry II Ab, E-cry I Ac

(2) A-nematode, B- Cellular defence
C-cry I Ac, D-cry II Ab, E-cry I Ab

(3) A-insects, B-cellular defense,
C-cry II Ab, D-cry I Ac, E-cry I Ab

(4) A-virus, B-protection against viral infection,
C-cry I Ac, D-cry II Ab, E-cry II Ac

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	1	2	1	2	1	3	2	3	1	2	1	3	2	1	3	2	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	4	2	3	3	1	2	4	2	4	1	1	3	1	3	1	2	1	2
Q.	41	42	43	44	45															
A.	1	1	3	4	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	4	3	2	1	3	3	4	3	3	4	2	1	2	4	3	3	3	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	4	1	3	4	1	3	4	2	4	4	1	2	3	3	1	4	4	3	1
Q.	86	87	88	89	90															
A.	2	3	1	1	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	1	1	3	2	4	4	3	4	3	2	4	4	1	2	2	4	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	1	2	3	4	2	4	2	4	4	4	3	2	3	3	2	4	3	1	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	4	1	2	2	1	4	3	2	2	3	2	3	2	4	2	1	4	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	2	4	4	4	2	4	1	4	4	4	2	1	1	3	3	1	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	3	1	3	3	3	3	4	2										

SOLUTIONS

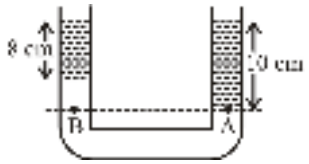
PHYSICS

1) $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$\frac{1}{v} - \frac{2}{-15} = \frac{1-2}{-10}$

⇒ v = – 30 cm

2) Mass of steam condensed = 425 g–(150 + 250)g = 25 g
Heat lost by steam = Heat gained by water & calorimeter
Let latent heat = L,
Heat released by steam = 25(L + 25)
Heat gained by water = Q_1 = (250)(1)(55) = 13750 cal
Heat gained by calorimeter = Q_2 = (150)(0.1)(55) = 825 cal
25(L + 25) = 13750 + 825
⇒ L + 25 = 583 ⇒ L = 558 cal g^{-1}

3) 

$P_A = P_B$

$10 \times \rho_x \times g = 8\rho_y g + 2\rho_{Hg} g$

$10 \times 3.36 = 8\rho_y + 2 \times 13.6$

$\rho_y = 0.8\ g/cm^3$

4) $\theta = \frac{\omega_i + \omega_f}{2} t$

$\theta = \frac{20\pi + 40\pi}{2} \times 10 = 30\pi \times 10 = 300\pi$ rad

Number of revolutions $N = \frac{300\pi}{2\pi} = 150$

5) $w_{friction} = \Delta KE + \Delta U$

$-\mu mg(\square + x) = -\frac{1}{2}mu^2 + \frac{1}{2}kx^2$

$-0.1 \times 50 \times (10.5) = \frac{1}{2}(5)(20)^2 + \frac{1}{2}K(0.5)^2$

$\frac{1}{8}K = 1000 - 52.5 = 947.5$

$\boxed{K = 7,580\ N/m}$

6) Time period for half part :

$T = 2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{1}{g}} = \frac{2\pi}{\pi} = 2\ \text{sec.}$

So 2° part will be covered in a time

$t = \frac{T}{2} = 1\ \text{sec.}$

For the left 1° part :

$\theta = \theta_0 \sin(\omega t)$

$1° = 2° \sin\left(\frac{2\pi}{T} \times t\right) \Rightarrow \frac{1}{2} = \sin\left(\frac{2\pi}{2} \times t\right)$

$\Rightarrow \frac{\pi}{6} = \pi \times t \qquad \Rightarrow t = 1/6$ sec.

Total time $\Rightarrow \frac{T}{2} + 2t$

$\Rightarrow 1 + 2 \times \frac{1}{6} = 1 + \frac{1}{3} = \frac{4}{3}$ sec.

7)

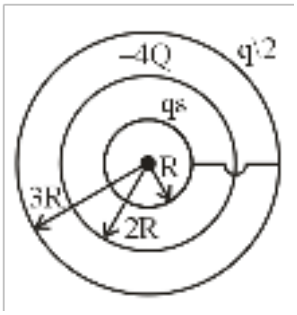
Though drift speed of electrons in metals is small, despite this, a large current can be obtained in a metal due to immense density of free electrons (10^{28} to 10^{29} m^{-3}) ⇒ Assertion is correct.
At room temperature, thermal speed of electrons are very high as electrons possess very small mass and according to KTG :

$KE = \frac{3}{2}K_BT = \frac{1}{2}m_eV_{th}^2$

Thus even small amount of thermal energy (K_BT) at room temperature (300 K) contribute very high speeds of electrons ⇒ Reason is correct.
Analysing above, both assertion and reason are correct but reason is not the correct explanation of assertion.

8)

Let after redistribution charge on inner most shell is q_1 and outermost shell is q_2. Hence.



$q_1 + q_2 = 2Q + 7Q = 9Q$ ...(1)

(By conservation of charge)

and $V_R = V_{3R}$

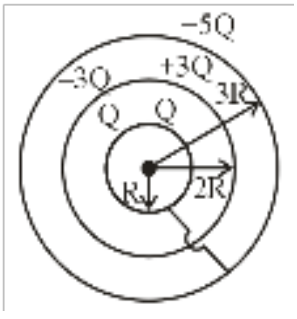
$\left[\frac{Kq_1}{R} - \frac{4KQ}{2R} + \frac{Kq_2}{3R}\right] = K\left[\frac{q_1 - 4Q + q_2}{3R}\right]$

$q_1 - 2Q + \frac{q_2}{3} = \frac{q_1 - 4Q + q_2}{3}$ ...(2)

Solving (1) & (2) final changes after redistribution

$q_1 = Q$, $q_2 = 8Q$

So, For shell $q_{inner} = 3Q$



$q_{outer} = 5Q$

9) $B = \mu_0 nI$

$= (4\pi \times 10^{-7})(200 \times 10^{-2}) \times 1.5$

$= 3.8 \times 10^{-2}$ Wb/m^2

Magnetic flux through each turn of the coil,

$\phi = BA$

$= (3.8 \times 10^{-2})(3.14 \times 10^{-4})$

$= 1.2 \times 10^{-5}$ weber

When the current in the solenoid is reversed, the change in magnetic flux = $2\phi = 2BA$

$= 2 \times (1.2 \times 10^{-5}) = 2.4 \times 10^{-5}$ weber

Iduced emf = $N(d\phi/dt)$

$$= 100 \times \left(\frac{2.4 \times 10^{-5}}{0.05}\right) = 0.048 \text{ V.}$$

10) For calculation of moment of inertia, the spatial distribution of mass respect to axis of rotation is considered, which mathematically comes as $I = Mk^2$, as whole of mass of body is considered to be situated at radius of gyration (not at COM). Hence assertion is incorrect. Moment of inertia of a rigid body (having definite shape and size) is not always zero but it is minimum at COM and depends upon orientation of axis also. So reason is also incorrect.

11) Let equation of a plane progressive harmonic wave $y = A \sin(\omega t - kx)$

wave speed $v = \frac{\omega}{k}$ and maximum particle speed $v_p = A\omega$

$v_p < v$ or $A\omega < \frac{\omega}{k}$

or $A < \frac{1}{k}$ or $A < \frac{\lambda}{2\pi}$

12) $F = F_R + F_a = \frac{2P_R}{C} + \frac{P_A}{C} = 2 \times \frac{0.3P}{C} + \frac{0.7P}{C}$

$F = \frac{1.3P}{C}$

13)

Initial radius $R = \frac{D}{2}$

initial surface area.

$$A_i = 4\pi R^2 = 4\pi\left(\frac{D}{2}\right)^2 = \pi D^2$$

Volume conservation :

$$\frac{4}{3}\pi R^3 = 27 \times \frac{4}{3}\pi r^3 \Rightarrow r = \frac{R}{3} = \frac{D}{6}$$

Surface area of one small drop

$$A_{one} = 4\pi r^2 = 4\pi\left(\frac{D}{6}\right)^2 = \frac{\pi D^2}{9}$$

Total final area :

$$A_f = 27 \times \frac{\pi D^2}{9} = 3\pi D^2$$

Increase in area

$\Delta A = A_f - A_i = 3\pi D^2 - \pi D^2 = 2\pi D^2$

Change in surface energy:

$\Delta E = T\Delta A = 2\pi TD^2$

14) Measured value of current

$$i = \frac{E}{\frac{R}{10} + \frac{(10R)(R)}{11R}} = 0.99\frac{E}{R}$$

Magnitude of % error in current

$$= \frac{1-0.99}{1} \times 100 = 1\%$$

True voltage across R is E.

Measured voltage = $\frac{i(10R)(R)}{11R}$

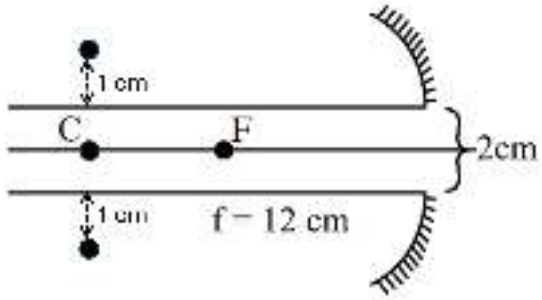
$$= 0.99\frac{E}{R} \times \frac{10R}{11} = 0.9E$$

% error in measurement = 10 %

15)

Object distance u = – 2f = – 24 cm (since C = 2f)

Separation in images = magnification × separation in parts



Object is at C, so image is at also C and of same size (real and inverted) for both parts. The two parts are separated vertically by 2 cm (distance between upper and lower part centers) The image distance for each part forms at the same object distance but shifted vertically due to the 2 cm separation.

Using magnification m = –1 (image inverted and same size)

Vertical separation between images = magnification × vertical separation of parts = |–1| × 2 cm = 2 cm

But here, actual lateral displacement equals twice the separation due to symmetry.

Reconsidering, when mirror is divided and object is on axis at C, vertical shift in images formed is twice the separation of mirrors, so:
Separation between images = 2 + 2 cm = 4 cm

16)

E_{eq} = (10 + 5) = 15V and
r_{eq} = (3 + 2) = 5Ω
For maximum power in R
R = r_{eq} = 5Ω
So current

$$I = \frac{E_{eq}}{R + r_{eq}} = \frac{15}{5+5} = 1.5A$$

Power in resistance R
$P = I^2R = (1.5)^2 \times 5W = 11.25$ W

17)

Energy of an EM wave is proportional to its frequency: E = hν. Microwaves have lower frequency than optical waves, so their energy is smaller.
However, all electromagnetic waves travel at the same speed in vacuum ($c \approx 3 \times 10^8$ m/s), so the statement “microwaves move faster than optical waves” is incorrect.

18)

After 1st collision, speed = (2 × 5) + 3 = 13 m/s
After 2nd collision, speed = (2 × 3) + 13 = 19 m/s
Final speed = 19 m/s

19) The square of the time period of a satellite is directly proportional to the cube of its orbital radius ($T^2 \propto R^3$). For small changes in radius, we use the error/percentage approximation method.
Relation: $T \propto R^{3/2}$
Percentage Change Formula: $\frac{\Delta T}{T} \times 100 = \frac{3}{2}(\frac{\Delta R}{R} \times 100)$
Given: $R_1 = R$ and $R_2 = 1.01R$. So, the change in radius $(\Delta R)$is $0.01R$, which is a 1% increase.

$$\frac{3}{2} \times 1\% = 1.5\%$$

Since radius increased, the time period also increases.

20) (a) V↑, T = Constant , P↓
(b) P↑ V↑
(c) P = Constant, V↑
(d) P↑ , V = Constant

21)

Power P = 32kW = 32×10^3 W = 32×10^3 J / s
Energy per fission E_f = 200 MeV
1 eV = 1.6×10^{19} J ⇒ 1 MeV = 1.6×10^{-13} J
$E_f = 200 \times 1.6 \times 10^{-13}$ J = 3.2×10^{-11} J
Fission rate R = $\frac{P}{E_f}$

$$R = \frac{32 \times 10^3}{3.2 \times 10^{-11}} = 10^{15}$$ per second

22) $\eta = \frac{2r^2}{9v_T}(\rho_s - \rho_f)g$ (r ⇒ same)

$$\frac{\eta_2}{\eta_1} = \left(\frac{v_{T_1}}{v_{T_2}}\right)\left(\frac{\rho_s - \rho_{f_2}}{\rho_s - \rho_{f_1}}\right)$$

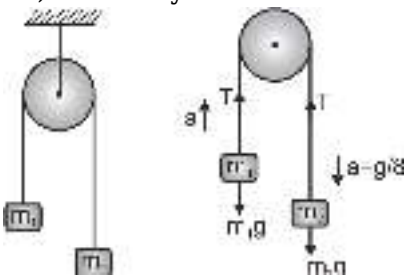
$$\Rightarrow \frac{\eta}{0.1} = \left(\frac{2.6}{1.5}\right)\left(\frac{2500-1000}{2500-1200}\right)$$

$$\eta = 0.1 \times \frac{26}{15} \times \frac{15}{13}$$

= 0.1 × 2 = 0.2 Pa-s

23) $E_1 = \phi_0 + 4KE_{max}$ ..... (1)
$E_2 = \phi_0 + KE_{max}$ ..... (2)
By solving 1 & 2
$\phi_0 = 4 \times 10^{-19}$ J

24) FBD of system



Considring $m_2 > m_1$
$m_2g - T = m_2a$ .....(i)
$T - m_1g = m_1a$ .....(ii)

$$a = \frac{(m_2 - m_1)}{m_1 + m_2}g$$

Given , a = g/8 ⇒ $\frac{g}{8} = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g$

$m_1 + m_2 = 8(m_2 - m_1)$
$7m_2 = 9m_1$

$$\frac{m_2}{m_1} = \frac{9}{7}$$

25)

1.6001.6001.600

26) $\vec{F} = e(\vec{v} \times \vec{B}) = -1.6 \times 10^{-19}((3.2\hat{i} + \hat{j}) \times 2\hat{k})$

$= -1.6 \times 10^{-19}\left[(3.2\hat{i} + \hat{j}) \times 2\hat{k}\right]$

$= 1.6 \times 10^{-19}\left[6.4\hat{j} - 2\hat{i}\right]$

Force directed along : {$F_x \Rightarrow$ –ve, $F_y \Rightarrow$ +ve}
Hence circular motion in x-y plane.

27) $S = ut + \frac{1}{2}at^2$

= 10 × 11 – 5 × 11 × 11
= 110 – 605 = –495 m

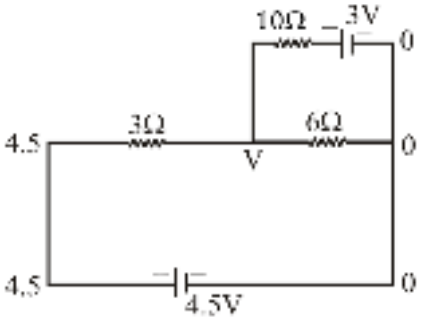
28) $r_n = r_0\frac{n^2}{z} = (0.53\text{Å})\frac{n^2}{z}$

$\Rightarrow n_1 = 2, \quad n_2 = 3$

$\Rightarrow \Delta E = [-1.51 - (-3.4)]\,\text{eV} = 1.89$ eV

Therfore it absorbed a photon of energy 1.89 eV

29)



$\frac{V-4.5}{3} + \frac{V-0}{6} + \frac{V-3}{10} = 0$

10V – 45 + 5V + 3V – 9 = 0
18 V = 54 V = 3

$i_1 = \frac{4.5-3}{3} = \frac{1}{2}A$

$i_2 = \frac{3-0}{6} = \frac{1}{2}A$

$i_3 = \frac{3-3}{10} = 0A$

30)

In series

$\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{1}{K_2} \Rightarrow K_{eq} = \frac{K}{2}$

$f = \frac{1}{2\pi}\sqrt{\frac{K_{eq}}{M}}$

$= \frac{1}{2\pi}\sqrt{\frac{K}{2M}}$

31) $f_o = \sqrt{f_1 f_2}$ & $Q = \frac{f_o}{f_1 - f_2}$

$f_o = \sqrt{100 \times 120} \approx 110 MHz$

Now $Q = \frac{f_o}{f_1 - f_2} = \frac{110}{120 - 100} = \frac{110}{20}$

$Q = 5.5$

32) $\frac{W}{Q} = \frac{nR\Delta T}{nC_p\Delta T} = \frac{32}{80} = \frac{2}{5}$

$\frac{R}{C_P} = \frac{2}{5}$

$5R = 2C_p = 2[C_v + R]$

$C_v = \frac{3R}{2} \Rightarrow f = 3$

33) $\frac{dQ}{dt} = \sigma eA\left(T^4 - T_0^4\right) = ms\frac{d\theta}{dt}$

so $\frac{dQ}{dt}$ will be same as 'e' is same but 's' is different so $\frac{d\theta}{dt}$ will be different.

34)

Refractive index depends on optical density, not just mass density. Optical density is related to how much a material slows down light, which depends on electrical polarizability, not directly on mass.

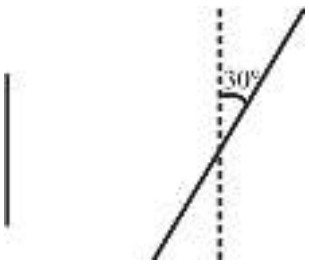
Refractive index $\mu = \frac{c}{v}$

No direct relation: μ ∝ mass density

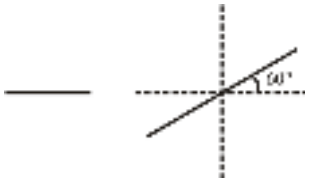
Turpentine has higher refractive index (~ 1.47) than water (~ 1.33) → Assertion is True

Mass density of turpentine (~ 870 kg/m^3) is less than that of water (~ 1000 kg/m^3) → Reason is False

35)



A $I_A\cos^2 30° = \frac{3I_A}{4}$



B $I_B\cos^2 60° = \frac{I_B}{4}$

$\frac{3I_A}{4} = \frac{I_B}{4}$

$\frac{I_A}{I_B} = \frac{1}{3}$

36)

Charge on outer surface of A & C plates

$= \frac{Q - 5Q + 10Q}{2} = \frac{6Q}{2} = 3Q$

Now,



37) – 7.5 cm– 7.5 cm

38) $\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2}$

$T^2 = t_1^2 + t_2^2$

39)

$J = \sigma E$

$\sigma = (n_e e\mu_e + n_h e\mu_h)$ $[n_e = n_h]$

$\sigma = n_e[\mu_e + \mu_h]e$

$\sigma = (1.072 \times 10^{10})\ (1.6 \times 10^{-19})\ [1350 + 480]$

$\sigma = 3.14 \times 10^{-6}$ mho/cm

40)

Normal shift towards point of rotation.

41)

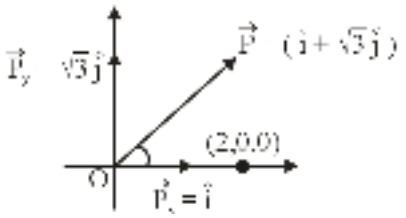
Energy dissipated = VAnt

$= \left(\frac{10}{7500}\right)$ (250) (50) (60 × 60)

$= 6 \times 10^4$ J

42) U = –MBcosθ and for unstable equilibrium
θ = 180º U = MB = 0.6 × 0.32 = 0.192 J

43) X = a + b ⇒ Δx = Δa + Δb
Now $\frac{\Delta X}{X} \times 100\% = \frac{(\Delta a + \Delta b)}{a+b} \times 100\%$
$= \left(\frac{\Delta a}{a+b} + \frac{\Delta b}{a+b}\right) \times 100\%$

44) 

For point (2, 0, 0) -
Due to $\vec{P}_x \Rightarrow \vec{E}_x = \frac{2k\vec{P}_x}{2^3} = \frac{k\hat{i}}{4}$
Due to $\vec{P}_y \Rightarrow \vec{E}_y = \frac{-k\vec{P}_y}{2^3} = \frac{-\sqrt{3}k\hat{j}}{8}$
$\left|\vec{E}\right| \sqrt{E_x^2 + E_y^2} = \frac{k}{8}\sqrt{4+3} = \frac{\sqrt{7}k}{8}$

45) λ_1 = 23 cm ; for second overtone = 123 cm
λ = 123 – 23 = 100 cm
Also $\frac{\lambda}{4}$ = 23 + e
25 = 23 + e ⇒ e = 2cm

CHEMISTRY

46) $\lambda_{min} = \frac{1}{R \times 4\left(\frac{1}{4} - \frac{1}{\infty}\right)} = \frac{1}{R} = x$
and $\lambda_{max} = \frac{1}{R \times 9\left(\frac{1}{9} - \frac{1}{16}\right)} = \frac{16}{7R}$
 $\lambda_{max} = \frac{16}{7R} = \frac{16x}{7}$

47)

$$A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)}$$

Initial moles:	2	2	0
Equilibrium moles	2 – x	2 – x	2x

The K_c expression is $K_C = \frac{[C]^2}{[A][B]}$.

$$4 = \frac{(2x)^2}{(2-x)^2}$$

$$2 = \frac{2x}{2-x}$$

$$4 - 2x = 2x$$

$$4 = 4x \Longrightarrow x = 1$$

The equilibrium moles of C are 2x.
Moles of C = 2(1) = 2 moles.

48) i. $HCOONH_4$: Formed from weak acid (formic acid) and weak base (ammonium hydroxide). Use formula where both pKa and pKb are involved.
ii. NH_4NO_3: Formed from strong acid (nitric acid) and weak base (ammonium hydroxide). Use formula with pKb and log C, making it acidic.
iii. CH_3COOK: Formed from weak acid (acetic acid) and strong base (potassium hydroxide). Use formula with pKa and log C, making it basic.
iv. KCl: Formed from strong acid (hydrochloric acid) and strong base (potassium hydroxide). Does not hydrolyze, pH is 7.

49) $2H_2$ + O_2 → $2H_2O$
with 30 ml 15 ml of O_2
of H_2 reacted
 H_2 is limiting reagent

50)

1. $MnO_4^- + C_2O_4^{2-} \rightarrow Mn^{2+} + CO_2$ **:** Mn changes from +7 → +2, C from +3 → +4. Redox reaction.
2. $FeC_2O_4 \rightarrow Fe^{3+} + CO_2$: only Fe is oxidized +2 to +3 ,no reduction partner. **not Redox**.
3. $Zn + NO_3^- \rightarrow Zn^{2+} + NH_4^+$: Zn changes from 0 → +2, N from +5 → 3. Redox reaction.
4. $VO + Fe_2O_3 \rightarrow V_2O_5 + FeO$: V changes from +2 → +5, Fe from+3 → +2. Redox reaction.

51)

$H_2O(\square) \rightarrow H_2O(g)$
$\Delta H = \Delta E + \Delta ngRT$
$\Delta E = \Delta H - \Delta ngRT$

$$= 40.66 - 1 \times \frac{8.314}{1000} \times 373$$

$= 37.5$ kJ mol^{-1}

52)

$C(s) + O_2(g) \rightarrow CO_2(g) + x$ kJ .....(1)

$$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g) + y\,kJ \quad .....(2)$$

– – – –

---

(1) – (2) $C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g) + (x - y)$ kJ

$\Delta H = y - x$ kJ/mol

53) 1. **Indicator:** Phenolphthalein is suitable as its pH range (8.2-10) covers the basic equivalence point. Methyl orange (pH 3.1-4.4) is not.
2. **Color Change:** Phenolphthalein is colourless in the acid (in the flask) and turns pink when the solution becomes basic at the end point.

54) $k = \left(\frac{k_1.k_2}{k_3}\right)^{2/3}$

$$= \left[\frac{\left(A_1 \cdot e^{-\frac{Ea_1}{RT}}\right) . \left(A_2 \cdot e^{-\frac{Ea_2}{RT}}\right)}{\left(A_3 \cdot e^{-\frac{Ea_3}{RT}}\right)}\right]^{2/3}$$

$$= \left[A'.e^{-\frac{Ea_1}{RT}-\frac{Ea_2}{RT}+\frac{Ea_3}{RT}}\right]^{2/3}$$

$$= (A')^{2/3} . e^{\left[\frac{-Ea_1 - Ea_2 + Ea_3}{RT}\right]^{2/3}}$$

On comparison with Arhenius Eq :

$E_a = \frac{2}{3}[Ea_1 + Ea_2 - Ea_3]$

$\Rightarrow E_a = \frac{2}{3}[180 + 80 - 50] = 140$ kJ

55) As shown in curve activation energy for backward reaction in absence of catalyst is S + P.

56) $\alpha = \frac{12.8}{(42 + 288.42)} \times 100 = 3.87\%$.

57) $E_{cell} = E^o_{cell} - \frac{0.0591}{n} \log Q$

$$E_{cell} = 1.1 - \frac{0.0591}{2} \log \frac{[Zn^{+2}]}{[Cu^{+2}]}$$

(i) If $[Zn^{+2}] = [Cu^{+2}] = 1M \Rightarrow E_{cell} = 1.1$ V

(ii) If $[Zn^{+2}] = 0.1$ M and $[Cu^{+2}] = 1M \Rightarrow E_{cell} > 1.1$ V

(iii) If $[Zn^{+2}] = 1M$ and $[Cu^{+2}] = 0.1$ M $\Rightarrow E_{cell} < 1.1$ V

(iv) If $[Zn^{+2}] = [Cu^{+2}] = 0.1$ M $\Rightarrow E_{cell} = 1.1$ V

So correct order will be

$\boxed{E_2 > E_1 = E_4 > E_3}$

58)

$V_1 = 200$ ml, $M_1 = M/10 = 0.1$ M

$N = M \times n$

$N_1 = 0.1 \times 2 = 0.2$ N (H_2SO_4, n = 2)

$N_2 = 0.1$ N (NaOH)

For complete neutralization no of equivalent of acid and base are equal

Formula : $N_1 \times V_1 = N_2 \times V_2$

$0.2 \times 200 = 0.1 \times V_2$

$V_2 = \frac{40}{0.1} = 400$ ml

59) • 30% w/w means 30 g urea in 100 g solution.

• Density = 1.2g/ml → 100 g solution has volume :

$$V = \frac{100}{1.2} = 83.33mL = 0.0833L$$

• Moles of urea :

$$n = \frac{30}{60} = 0.5mol$$

• Molarity = 0.5 / 0.0833 ≈ 6 M

60)

$P_{ideal} = x_A \cdot P°_A + x_B \cdot P°_B$

• $P_{observed} > P_{ideal}$ → Positive deviation

• $P_{observed} < P_{ideal}$ → Negative deviation

• $P_{observed} = P_{ideal}$ → Ideal solution

$$x_A = \frac{1}{(1+2)} = \frac{1}{3} = 0.333$$

$$x_B = \frac{2}{3} = 0.667$$

$P_{ideal} = (0.333)(150) + (0.667)(300)$

$= 50 + 200 = 250$ torr

$P_{observed} = 240$ torr, which is less than ideal (250 torr).

Hence, solution shows negative deviation.

61)

(A) For d-block elements :

In a period very irregular pattern is observed by for these '3' consecutive elements.

Cr < Mn < Fe (IE order)

E.A. ⇒ O < Te < Se < S

(B) After 1st ionisation :

Electronic configuration :

$C^+ \rightarrow 1s^2\ 2s^2\ 2p^1$

$N^+ \rightarrow 1s^2\ 2s^2\ 2p^2$

$O^+ \rightarrow 1s^2\ 2s^2\ 2p^3$ Half filled, so IE (O^+) more

$F^+ \rightarrow 1s^2\ 2s^2\ 2p^4$

2nd IE order

O > F > N > C

I.E. order :

O^- IE , S^- IE

⇓ ⇓

O (EA) < S (EA)

IE : $O^- < S^-$

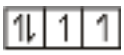
62) (P) W(2, 8, 6)

Electronic configuration

⇒ $1s^2$, $2s^2$, $2p^6$, $3s^2\ 3p^4$

p-block element

16th group element



$3p^3$ So paramagnetic with two unpaired electron.

(Q) X(2, 8, 18, 8)

E.C. ⇒ $1s^2$, $2s^2\ 2p^6$, $3s^2\ 3p^6$, $4s^2$, $3d^{10}$, $4p^6$

p-block element → Krypton

18th group element

Inert gas

(R) Y(2, 8, 18, 25, 8, 2)

E.C. ($_{63}$Eu) ⇒ $_{54}$[Xe] $6s^2\ 4f^7$

f-block element

(S) Z(2, 8, 18, 18)

E.C. ⇒ $_{36}$[Kr] $5s^0\ 4d^{10}$

Z ⇒ 46 ⇒ 10th group element

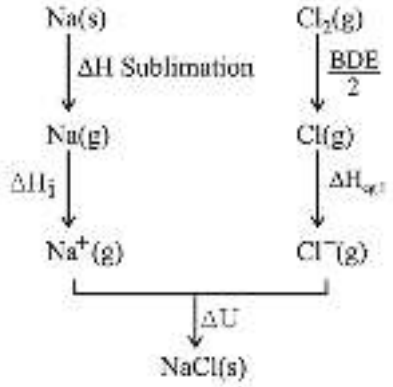
Ans. (C) P→1; Q→4; R→2; S→3.

63) NCERT Pg. # 111

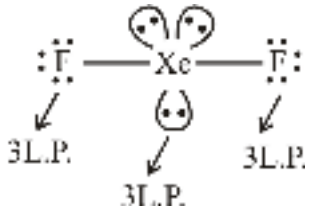
64)

Solution: Four species do not exist: Be_2 , Ne_2 (Bond Order = 0); BF_6^{3-} (Boron lacks d-orbitals for hypervalency); and PH_5 (unstable hydride).

Final Answer: Option (3)



65)



66)

Total lone pairs in XeF_2 = 9

67)

$CoBr_2Cl.4NH_3$

Co^{+3} C.N. = 6

$[Co(NH_3)_4Br_2]Cl$

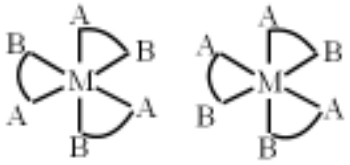


$[Co(NH_3)_4Br_2]^+ \ Cl^-$

with $AgNO_3$ (Excess) → (with complex) AgCl

68)

M $(AB)_3$



Both are optically active

$[Ni(CO)_4]$ bonding can be explained by M.O.T. $KMnO_4$ color due to L → M charge transfer. (LMCT)

Co^{+3} act 3 as L.A

69) Conceptual (Synergic bonding)

70)

CrO - Basic

Cr_2O_3 - Amphoteric

71)

SO_2 gas decolourize acidified $kMnO_4$ solution

$kMnO_4 + SO_2 \xrightarrow{H^+} Mn^{+2} + SO_4^{-2} + H_2O$

72)

NCERT XII # 201

73) **Concept-**

1. Naturally occurring carbon contains 2 stable isotopes, ^{12}C, ^{13}C

This is correct. Carbon naturally exists as a mixture of ^{12}C and ^{13}C, both of which are stable

2. ^{14}C is a radioactive isotope with half life 5770 years.

This is also correct. ^{14}C is a radioactive isotope used in radiocarbon dating, and its half-life is 5770 yrs,

3. fact

4. All of the above

Since all the individual statements are correct, this is the correct answer.

Therefore, the correct answer is 4. All of the above.

Answer-Option(4)

74)

Each analytical group of cations is precipitated by a specific reagent:

A. Group II : $Cu^{2+} \rightarrow H_2S$ + dil. HCl
B. Group III : Al^{3+}, $Fe^{3+} \rightarrow NH_4OH + NH_4Cl$
C. Group IV : Co^{2+}, $Ni^{2+} \rightarrow H_2S + NH_4OH$
D. Group V : Ba^{2+}, $Ca^{2+} \rightarrow (NH_4)_2CO_3 + NH_4OH$

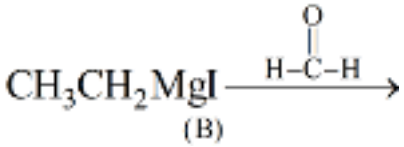
Solution:

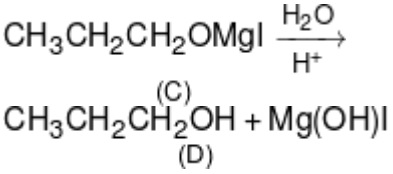
A. **A.** $Cu^{2+} \rightarrow$ i. H_2S + HCl
B. B. Al^{3+}, $Fe^{3+} \rightarrow$ iii. $NH_4OH + NH_4Cl$
C. C. Co^{2+}, $Ni^{2+} \rightarrow$ iv. $H_2S + NH_4OH$
D. D. Ba^{2+}, $Ca^{2+} \rightarrow$ ii. $(NH_4)_2CO_3 + NH_4OH$

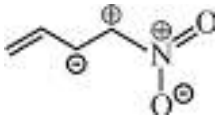
75)

NH_3 gas brown colour ppt with Nesseler reagent known as Iodide of millon's base.

76) $CH_3CH_2OH \xrightarrow{P+I_2} \underset{(A)}{CH_3CH_2I} \xrightarrow[ether]{Mg}$

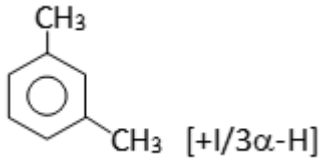


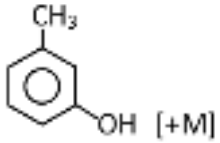


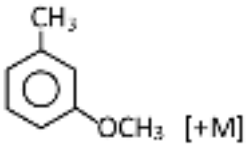
77) 

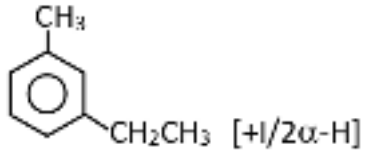
Least stable due to closer same charges

78) **Solution :**

1. 

2. 

3. 

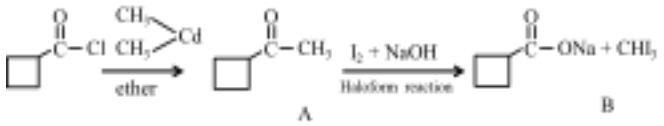
4. 

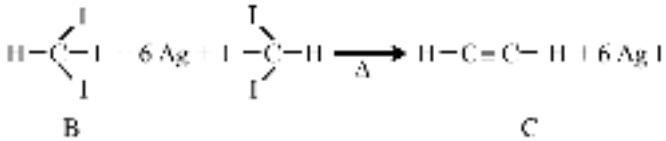
+M of –OH > $-OCH_3$

 order of rate of ESR is : 2 > 3 > 1 > 4

Final Answer option (2)

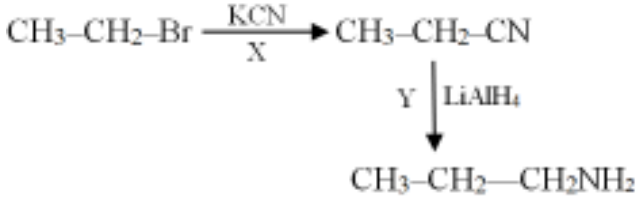
79) **Solution:**

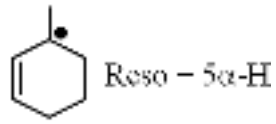
(1) 

(2) 

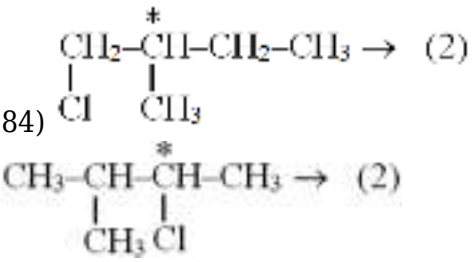
Final Answer: Option (3)

80) Lactose & fructose gives positive tollen's test

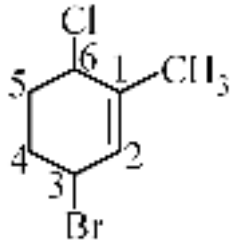
81) 

82) 

83) NCERT XII Pg. # 294
In T.S. C is sp^2 hybridsed.

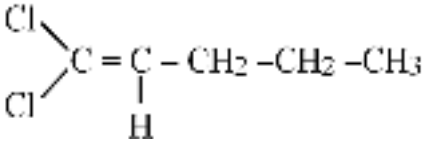
84) 

Total chiral compounds are 4

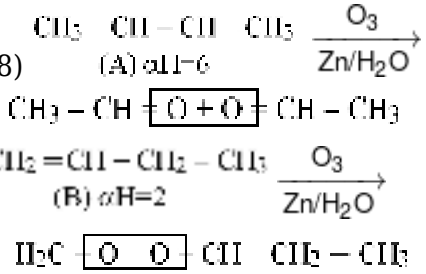
85) 

3–bromo–6–chloro–1-methyl cyclohexane.

86) 'N' converted to $(NH_4)_2SO_4$ → kjeldahl method.
Blue colour of 'N' in Lassaigne test
→ $Fe_4[Fe(CN)_6]_3$,
Compound formed in Lassaigne test of N
→ $Fe(SCN)_3$,
If S is also present
Voilet colour during → $Na_4[Fe(CN)_5NOS]$
Lassaigne test of S

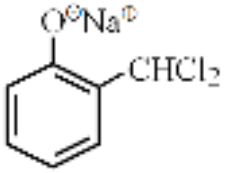
87) 

1, 1-Dichloropent-1-ene. Does not show GI

88) 

Stability :- A > B

89)

Intermediate of Riemer Tiemann Rx^{n} is 

90)

Gabriel phthalimide reaction is used for synthesis of primary aliphatic amines.

BIOLOGY

91)

NCERT-XI Pg. # 6, 7 & 8

92) NCERT XI Pg.- 6

93) NCERT XI, Pg. No. # 13

94) NCERT XI, Pg. No. # 16, 18

95) NCERT, Pg. # 28

96)

NCERT-XI Pg. # 29

97) NCERT Pg. # 32

98)

NCERT-XI Pg. # 65, 66

99)

NCERT-XI Pg. # 76, 77

100) NCERT-XI, Pg. # 76, 77

101)

NCERT Pg. # 94, 95, 96

102)

NCERT-XI Pg. # 95, 98

103)

NCERT-XI, Pg. # 121-124

104)

NCERT XI, Page # 166, Fig. 10.2

105)

NCERT-XI Pg. # 126, 127

106)

NCERT-XI Pg. # 112, 117

107) NCERT XI Pg. 144

108) NCERT XI Pg # 146

109) NCERT XI, Pg No # 231

110)

NCERT-XI, Pg. # 157

111) NCERT-XI, Pg. # 247 (2nd Para), 248 (2nd last Para), 249 (2nd Para), 250 (1st Para & 2nd last Para) (E)
NCERT-XI, Pg.# 247 (दूसरा पेराग्राफ), 248 (आखिरी पेराग्राफ), 249 (तीसरा पेराग्राफ), 250 (पहला पेराग्राफ), 251 (पहला पेराग्राफ) (H)

112) NCERT-XI, Pg. # 246

113) NCERT 11th Class : 11.6.3 Chemiosmotic Hypothesis and 12.4.2
Electron Transport System (ETS) and Oxidative Phosphorylation

114)

NCERT XI Pg. # 211

115)

NCERT-XI Pg. # 135

116)

NCERT-XII Pg. # 9, 11

117)

NCERT-XII Pg. # 10, 18, 19

118) NCERT-XII, Pg. No. # 189

119) NCERT Pg. # 199

120)

NCERT XII, Pg # 191, 193, 196

121) NCERT Pg. # 210

122) NCERT Pg. # 258

123) NCERT Pg. No. # 220-221

124) NCERT Pg. # 222, 223

125) NCERT (XII) ; Page No. # 89

126) NCERT (XII) ; Page No. # 80

127) NCERT-XII, Pg # 101

128) NCERT Pg. # 92, 93, 94, 99

129) NCERT, Pg. # 81/90

130)

NCERT-XII Pg. # 81

131) NCERT XII Pg. No. # 73

132) XII NCERT Pg. 60, 61

133) NCERT Page No. #70

134) NCERT Page-94

135)

NCERT XII Pg # 79

136) NCERT XI Page No. # 40

137) NCERT-XI, Pg # 48

138) NCERT XI Pg # 42

139) NCERT XI Pg. No. 40-44

140) NCERT-XI, Pg. # 44-45

141) Old NCERT

Module

142)

NCERT_XII_Pg. No. 114

143) NCERT_Pg. No. # 113

144) NCERT_Pg. No. 81

145) NCERT XI Pg # 82

146) NCERT Class-XII, Page No. 193, 194

147) NCERT Pg. # 203

148) NCERT XIth Pg.# 195.

149) NCERT Pg. # 185

150) NCERT Pg. # 187, 188

151) NCERT Pg. # 272

152) NCERT (XII) ; Page No. # 321 (E)
NCERT (XII) ; Page No. # 320 (H)

153) NCERT Pg. No. # 232, 233, 234

154) NCERT (XI) ; Page No. 243

155) NCERT XI Pg. # 248

156) NCERT XI Pg.#240,241

157) NCERT XI Pg.#242

158) NCERT Pg. No. # 227

159) NCERT Pg. # 227

160) NCERT, Pg # 34

161) After ovulation there are always 14 days (fixed phase), so total number of days in menstrual cycle will be 17 + 14 = 31 days

162) NCERT Pg. # Module

163) NCERT XII Pg. No. # 28, 30

164) NCERT(XII) Page#44/50(H) Para:3.2

165) NCERT-XII, Pg. # 145

166) NCERT Pg.# 130-134
Bacterial (Pneumonia, typhoid, TB, Diphtheria, Plague)

167) NCERT Pg. # 137

168) NCERT XII Pg. No. # 136

169)

NCERT XI Page No. # 44

170)

Question Explanation: The question asks where maximum and minimum reabsorption occurs in the nephron.

Concept: Selective Tubular Reabsorption

Solution:

A. Maximum reabsorption → PCT → Site of maximum reabsorption (~65–70% of filtrate, including water, ions, glucose, amino acids).

B. Minimum reabsorption → Ascending limb of Loop of Henle → The ascending limb is impermeable to water, and only permeable to some amount of salt to maintain osmolarity gradient of medulla.

Final answer: Option 3

171) NCERT XI (E), Page # 291

172) NCERT XII Page No. 133 Pare No. 7.4

173) NCERT Based
Final answer : 3

174)

NCERT-XIIth Pg. # 37

175) NCERT XII Pg # 157

176)

NCERT XII, Pg # 182

177) NCERT XII Pg. # 170-171

178) NCERT Pg. # 179,181

179)

NCERT XII Page No. # 183-184

180)

NCERT-XII Pg # 179-181